

The Manual adheres to the scope and sequence of most one major course nationwide also strives to make the PRACTICAL MANUAL OF DIVERSITY AND FUNCTION OF INVERTEBRATES&CHORDATS (ALLIED ZOOLOGY PRACTICAL-I) (A Revised Syllabus based on Periyar University) as a discipline, interesting and accessible to students. In addition to a comprehensive coverage of core concepts and foundational invertebrates & chordates and its importance features that draw learners into the discipline in meaningful ways. This Manual provides a fundamental concept of invertebrates & chordates with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful book on the text book of invertebrates & chordates.



MR.S.KALAIMANI.,M.Sc.,M.Phil.,SET.,M.Ed.,SET., is an 17 years of Teaching Experience, He Expert in Biological Research in various fields of Biological Studies, who has currently working as a Assistant Professor of Zoology, P.G and Research Department of Zoology, J.K.K.Nataraja, College of Arts & Science, Kumarapalayam, Namakkal District.



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SUBRAMANI, KRISHNAMOORTHY

KALAIMANI SUBRAMANI
SHENKANI KRISHNAMOORTHY

PRACTICAL MANUAL OF ALLIED ZOOLOGY

ALLIED ZOOLOGY PRACTICAL -I
(A Revised Syllabus based on Periyar University)

 **LAMBERT**
Academic Publishing

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Imprint

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PRACTICAL MANUAL OF DIVERSITY AND FUNCTION OF INVERTEBRATES&CHORDATS

ALLIED ZOOLOGY PRACTICAL -I

(A Revised Syllabus based on Periyar University)

Edition-2024

Authors:

Dr. S. KALAIMANI.,M.Sc.,M.Phil.,M.Ed.,SET(L.sci),,SET(E.du),,DCA.,

Assistant Professor

PG and Research Department of Zoology

J.K.K. Nataraja College of Arts & Science

Kumarapalayam, Namakkal D.t, Tamilnadu, India.

Dr. K.SHENKANI.,M.Sc.,M.Phil.,SET (L.sci),,Ph.D

Assistant Professor

PG and Research Department of Zoology

J.K.K. Nataraja College of Arts & Science, Kumarapalayam,

Namakkal D.t,Tamilnadu, India.

FOREWORD

The Manual adheres to the scope and sequence of most one major course nationwide also strives to make the **PRACTICAL MANUAL OF DIVERSITY AND FUNCTION OF INVERTEBRATES&CHORDATS (ALLIED ZOOLOGY PRACTICAL-I)** (A Revised Syllabus based on Periyar University) as a discipline, interesting and accessible to students. In addition to a comprehensive coverage of core concepts and foundational invertebrates & chordates and its importance features that draw learners into the discipline in meaningful ways. This Manual provides a fundamental concept of invertebrates & chordates with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful book on the text book of invertebrates & chordates. I am appreciating the authors to chosen this Manual on basic preparatory knowledge of invertebrates & chordates. on the aspect include number of Manual are available in the form of hand book, question bank for competitive examinations etc. except from this for the student community limited numbers of this like books are available.UG or PG level lack of syllabus covered text books are available, this book might be useful to basic and advanced learner of invertebrates & chordates. Written of this Manual contain simple language and easy to understand a basic learner. This Manual offers a valuable guide to the intellectual. This book can help teachers develop a shared vision and understanding of basic concepts of invertebrates & chordates and techniques of Nutrition& Diet and its flexible uses across subjects and grade levels. This Manual has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of academic scholarship. Instructors and students alike will find that this Manual offers a strong introduction to invertebrates & chordatas in an accessible format.

X. Venci Candida

Dr. X. Venci Candida,
Assistant Professor of Zoology,
Holy Cross College (Autonomous),
Nagercoil

PREFACE

UG and PG Zoology students, who are looking for a quick revision before the A Students who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her this difficulty is mainly due to understand standard references. While preparing this Manual “**PRACTICAL MANUAL OF DIVERSITY AND FUNCTION OF INVERTEBRATES&CHORDATS (ALLIED ZOOLOGY PRACTICAL-I)**” (A Revised Syllabus based on Periyar University) has been prepared to serve the special purpose of college students. This Manual explains the basic Clinical Nutrition technique in a simplified manner. This Manual can be used for preparation semester examination of Periyar university affiliated colleges. This book support to the B.sc & M.sc- Zoology students to get confidence and also reference just before attending University Examination and other Competitive Examination. This Manual has been written to UG and PG Zoology Students of Life sciences, no such Manual which cover suitable original picture with needed explanation of invertebrates & chordates. This manual is useful for the preparation of state Universities and College integrated Certificate and Diploma courses field of life sciences.

Dr. S. KALAIMANI
Dr. K. SHENKANI

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My specific thanks to Dr. SHENKANI.,M.Sc.,M.Ed.,M.Phil.,SET(L.sci)., Ph.D.,Assistant Professor, PG and Research Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this book.

My specific thanks to Dr.S.V. Ramesh, M.com., MPhil.,Ph.D., Assistant professor, PG &Research department of commerce, J.K.K.Nataraja college of arts and science, Kumarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this book successfully.

I wish to express my profound thanks to All the Staff Members, Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal D.t.

Last but not least wish to extend my sincere thanks to all my friends and relatives for their co-operation and support in this book to make success.

Dr. S. KALAIMANI

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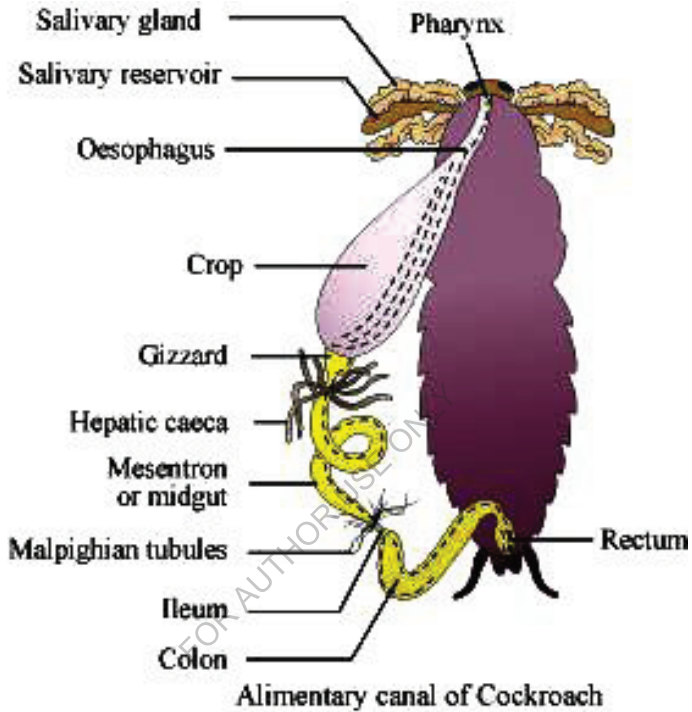
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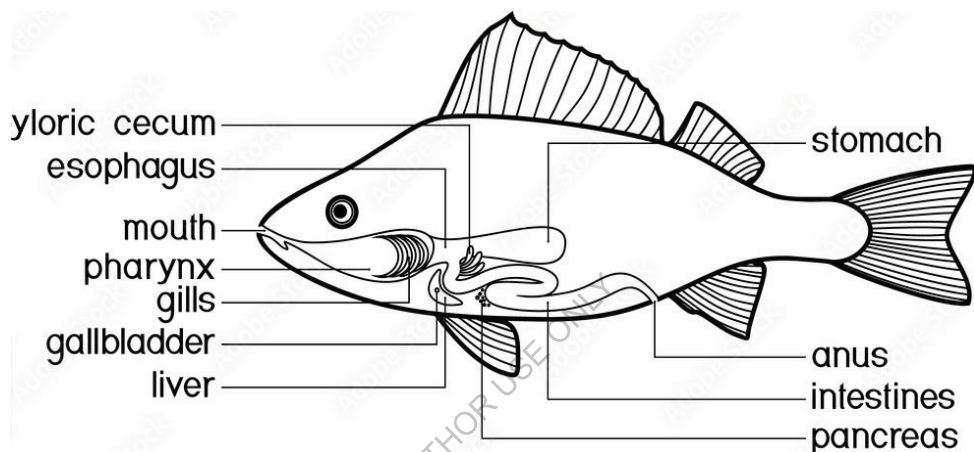
MAJOR PRACTICAL

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A. COCKROACH –DIGESTIVE SYSTEM



FISH- DIGESTIVE SYSTEM



QUALITATIVE DETECTION OF EXCRETOR

PRODUCTS

(AMMONIA, UREA & URIC ACID).

Aim

To perform a simple test to identify the presence of Ammonia, Urea and Uric acid in the given sample.

Material Required

- Spatula.
- Dropper.
- water sample from fish tank, Cows Urine sample and birds' droppings white portion only
- Nessler's Reagent Solution, Phenol red solution, Fehling's solution
- sodium carbonate solution
- Horse gram powder
- Measuring cylinder.

REAGENTS

- 2 mL of Nessler's Reagent Solution,
- Phenol red solution
- Folin's silver nitrate solution

PROCEDURE

TEST FOR AMMONIA

1. Take 2ml of given sample A solution in a clean test tube.
2. Add 3drops of 2% of Nessler's Reagent Solution to it.
3. Keep the solution in a boiling water bath for about 10 minutes.
4. If there is the formation of reddish-brown precipitate then
5. The presence of Ammonia is confirmed.

TEST FOR UREA

1. Take 2ml of given sample B solution in a clean test tube.
2. Add 3drops of **Phenol red solution** to it.
3. Add a pinch of horse gram powder
4. If there is the formation of Deep brown or red precipitate
5. The presence of Urea is confirmed.

TEST FOR URIC ACID

1. Take 2ml of given sample C (White portion of bird's droppings) solution in a clean test tube.
2. Add 3ml of saturated Sodium carbonate is added
3. Add 3 ml of Fehling's **solution** to it.
4. If there is the formation of Blue color.
5. The presence of Urea is confirmed.

OBSERVATION AND CONCLUSION

SAMPLE	AMMONIA	UREA	URIC ACID
SAMPLE A	+	-	-
SAMPLE B	-	+	-
SAMPLE C	-	-	+

RESULT:

1. The given sample a is AMMONIA
2. The given sample b is UREA
3. The given sample c is URIC ACID

MINOR PRACTICAL

MOUTH PARTS OF HONEY BEE, MOSQUITO

AIM

Mounting of mouth parts of, Honey bee, Mosquito

MATERIALS REQUIRED

Live house fly, honey bee, mosquito, forceps, test tube holder, small Petridis/watch glass, dropper,

PROCEDURE

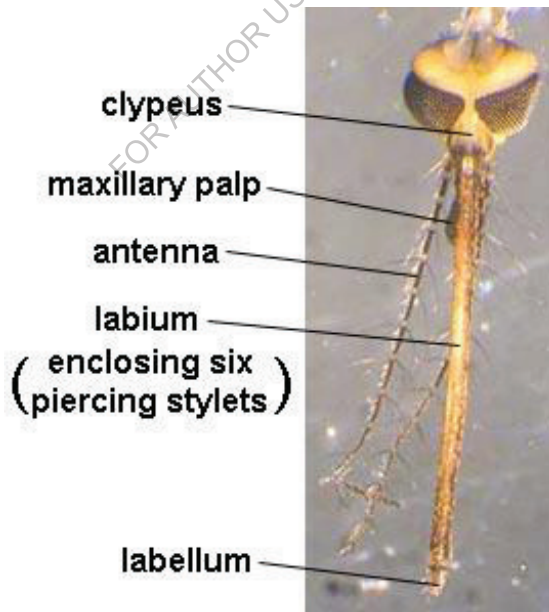
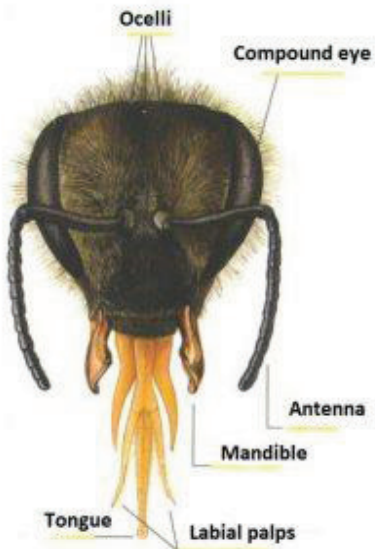
1. Put live honey bee into the beaker containing 10% alcohol, after narcotization of honey bee to cut piece of
2. The cut head portion of honey bee put into the test tube with help of forceps
3. To add 10% KOH in to the test tube with portion of mouth parts.
4. Boil the test tube up to 1:30 minutes help of spirit lamp
5. After boiling take piece of honey bee head portion on the clear slide help of dropper. Put cover glass.
6. Put under the compound microscope and observe mouth parts of honey bee appearance

RESULT

The focused structure consists of 3 region the mandible, maxilla, labial palp, proboscis and glossa.

PROCEDURE

1. Using forceps and/or micro dissection scissors, remove legs and wings to facilitate dissection of the abdomen.
2. Keep a small beaker or similar container filled with distilled water nearby for the purpose of rinsing instruments between dissections.
3. While restraining the bee with forceps, utilize the micro dissection scissors to cut laterally along the dorsal abdominal wall between the first and second tergites



FISH – CYCLOID SCALE, CTENOID SCALE AND PLACOID SCALE

- A scale is a small rigid plate that grows out of animal's skin to provide protection. Fish scales are dermal in origin.
- There are four types of scales in fishes, namely Cosmoid, Ganoid, Placoid, and Leptoid. Leptoid scales are found in most bony fishes.
- As they grow, they add concentric layers and are arranged so as to overlap in a head-to-tail direction, allowing a smoother flow of water over the body.
- These are of two types: Cycloid and Ctenoid.
- Cycloid scales have a smooth outer edge, and are most common on fish with soft fin rays, such as salmon and carp (Example: Rohu, Catla, Mrigel etc.).
- Ctenoid scales have a toothed outer edge, and are usually found on fish with spiny fin rays, such as bass and perches (Example: Bhetki, Koi, Tilapia etc.)
- Placoid Scales are found in cartilaginous fishes, like sharks. These scales, also called denticles, are microscopic, similar in structure to teeth and have one median spine and two lateral spines.

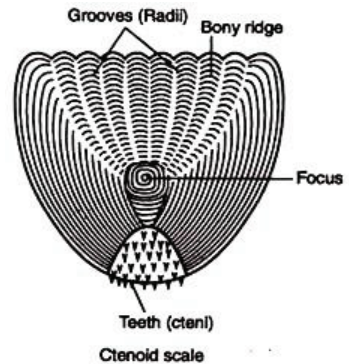
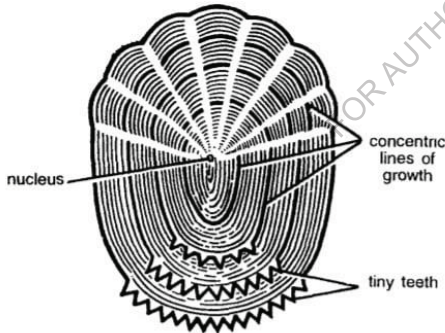
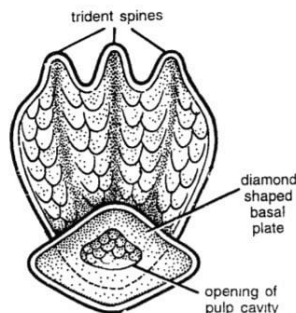


Fig. 3.3(d) : Ctenoid scale



AIM

Mounting of fish Cycloid scale, Ctenoid scale and Placoid scale

MATERIALS REQUIRED

Live fish, Forceps, Small Petridis/watch glass, Dropper, Clean glass slide, Compound microscope,

PROCEDURE

- i) A small piece of skin is cut from the dorsal surface of a shark (*Scoliodon sp*).
- ii) It is then put in a hard glass test tube containing 5 to 10% potassium Hydroxide (KOH) solution and is boiled with constant stirring till the skin dissolves.
- iii) The contents of the test tube are poured in a watch glass and are allowed to cool.
- iv) The scales settle at the bottom and the fluid is removed by decantation (if required, with water) till the last trace of KOH is removed.
- v) With the help of a pipette, little amount (preferable one drop) of fluid with the scales are put on a slide.
- vi) The excess water is removed with a piece of blotting paper, stained, if required and is mount in glycerin, followed by microscopic observation.

PREPARATION OF PLACOID, CYCLOID & CTENOID SCALES

- i) A few scales is removed from the fish. For cycloid, the fish may be a carp, but for ctenoid, it may be a perch like Koi or Tilapia.
- ii) The scales are put in a watch glass containing 10% KOH solution.
- iii) The scales are stirred slowly with a needle or brush till the covering epithelium dissolves.
- iv) The scales are now washed thoroughly with water to remove the last trace of KOH, followed by staining, mounting in glycerin and microscopic observation.

RESULT

The focused structure consists of stained cyloid, ctenoidand, placoid scale

ABO -BLOOD GROUP

AIM

Identification of blood groups

OBJECTIVE

To determine the blood group and Rh factor of an individual.

MATERIAL REQUIRED

- Anti A Sera
- Anti B Sera
- Anti Rh D Sera
- Cavity slide
- Disposable Mixing Stick
- Blood Lancet
- 70% Alcohol/ Spirit
- Cotton

PROCEDURE

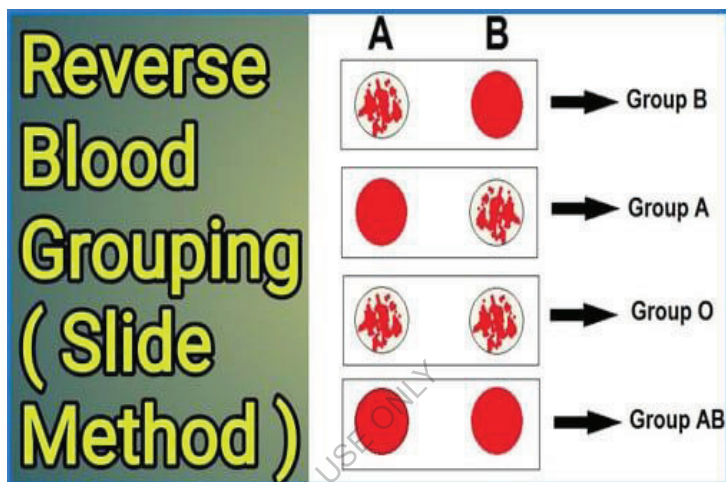
The Blood Grouping test can be done by two methods such as Slide or tile method and the Tube method.

SLIDE OR TILE METHOD

1. Dangle the hand down to increase the flow of blood in the fingers
2. Clean the fingertip to be pierced with spirit or 70% alcohol (usually ring or middle finger).
3. With the help of the sterile lancet, pierce the fingertip and place one drop of blood in each of the cavities.
4. Add one drop of antiserum into each cavity as shown below:
5. Mix each blood drop and the antiserum using a fresh mixing stick.
6. Observe agglutination in the form of fine red granules within 30 seconds.
Anti Rh D takes a slightly longer time to agglutinate compared to Anti A and Anti B.

RESULT OF ABO BLOOD GROUPING

SLIDE METHOD



RESULT OF ABO BLOOD GROUPING

S. No.	Anti A	Anti B	Anti RhD	Blood Group
Slide 1	Agglutination	No agglutination	Agglutination	A +ve
Slide 2	No agglutination	Agglutination	Agglutination	B +ve
Slide 3	Agglutination	Agglutination	Agglutination	AB +ve
Slide 4	No agglutination	No agglutination	Agglutination	O +ve

SPOTTERS

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AMOEBA

SYSTEMATIC POSITION

Phylum: Protozoa

Subphylum: Sarcomastigophora

Class: Rhizopoda

Order: Amoebida

Genus: *Amoeba*

Species: *proteus*

COMMENTS:

1. Animal is irregular shape, with simple or branched pseudopodia measuring 250 to 600 microns in diameter.
2. Cytoplasm is differentiated into ectoplasm and endoplasm. Ectoplasm contains ectoplasmic ridges.
3. Body of the animal is covered by a thin, delicate and permeable plasma membrane, called as plasma lemma.
4. Endoplasm contains nucleus, food vacuole, contractile vacuoles, water globules and crystals.
5. Permanent posterior end is called as uroid.
6. Withdrawal of pseudopodium and new pseudopodium containing endoplasm is present
7. Feeding may be studied by giving carmine. Nutrition is holozoic.
8. Reproduction by binary fission & multiple fission
9. *Amoeba proteus* move by the formation of Pseudopodia are blunt, fingerlike extensions of the ectoplasm containing endoplasm (lobo podium).

PARAMECIUM

SYSTEMATIC POSITION

Phylum ... Protozoa

Subphylum.... Ciliophora

Class ... Ciliata

Subclass... Euciliata

Order..... Holotricha

Suborder..... Trichostomata

Family.... Paramecidae

Genus..... *Paramecium*

COMMENTS:

- 1 Commonly called as slipper animalcule, being microscopic, elongated, slipper- shaped cigar-shaped or spindle shaped.
2. Most familiar and extensively studied protozoans.
3. Pellicle covers the body. It is clear firm and elastic cuticular membrane. Pellicle has series of polygonal or hexagonal depressions for trichocysts.
4. *Paramecia* propel themselves by whiplash movements of their cilia, which are arranged in tightly spaced rows around the outside of their body.
5. The beat of each cilium has two phases: a fast "effective stroke," during which the cilium is relatively stiff, followed by a slow "recovery stroke," during which the cilium curls loosely to one side and sweeps forward in a counter-clockwise fashion.

TRYPANOSOMA GAMBIANS

SYSTEMATIC POSITION:

Phylum : Protozoa

Subphylum: Sarcomasigophora

Super class: Mastigophora

Class : Zoomastigophora

Order : Kinetoplastida

Genus : *Trypanosoma*

Species : *Gambians*

COMMENTS

1. Unicellular, microscopic, simple animal commonly found in Africa.
2. It is an endoparasite found in blood and other tissues of man and other vertebrates.
3. It causes Trypanosomiasis or Sleeping sickness
4. It is digenetic i.e. it completes its life cycle in two hosts: principal host in man and

Intermediate host is a blood sucking insect *Glossina palpalis* or Tsetse fly.

5. It is a polymorphic species exists in man and a trichinidial forms in tsetse fly
 6. A single flagellum arises from a basal body placed near the posterior end
- Reproduction take place by longitudinal binary fission

EUGLENA

SYSTEMATIC POSITION:

Phylum ... Protozoa

Subphylum ... Plasmodroma

Class.... Mastigphora

Order... Euglenoidina

Genus... *Euglena*

COMMENTS:

1. Body of animal is simple fuse form, spindle shaped plump like red or green colour.
2. *Euglena* measure 50 to 100 microns in length.
3. Outer covering is called as pellicle which are marked by a spiral striation called as myonemes
4. Anterior end has funneled shaped cytosome which leads into a flask –shaped cytopharynx.
5. On one side is a red mass of hematochrome called stigmata which is photosensitive
6. Endoplasm contains nucleus, chloroplast and other ultra structural organelles
7. Nutrition holophytic or saprophytic and reproduction by longitudinal division or encystment.
8. *Euglena* does not take solid food but lives entirely by autotrophic and saprozoic nutrition.
9. It is unique animal with floral mode of nutrition and funnel mode of life and reproduction
10. *Euglena viridis*, *E. rubra*, *E. sanguine* and *E. fusiformis* are common species they respond to various stimuli such as light heat etc.

PLASMODIUM

SYSTEMATIC POSITION:

Phylum: Protozoa

Subphylum: Plasmodroma

Class: sporozoa

Subclss: Telosporidia

Oder: Haemosporidia

Genus: *Plasmodium*

COMMENTS:

1. *Plasmodium* is an intracellular blood parasite of man and other vertebrates and Causes malaria.
2. The life history of plasmodium (Fig.8.40) is completed in two hosts, viz., partly in Definitive, host the man and partly in intermediate host, the female anopheles mosquito.
3. When an infected female anopheles mosquito bites a man, sporozoites are introduced in the blood from where they reach in live cells through blood streams and multiply to form merozoites.
4. After a few schizogenous cycles in the liver, the merozoites enter the red blood Corpuscles (R.B.C) and feed on the contents of R.B.C.
5. After 2-8 schizogenous changes in the main blood stream, the merozoites assume different shapes and known as gametocytes.
6. Gametocytes cannot develop further in the blood of man; therefore, they wait for female anopheles' mosquito to suck them with the blood.
7. When gametocytes are sucked in by the female anopheles with the blood of man, they undergo sporogony for further development.

LEUCOSOLENIA

SYSTEMATIC POSITION:

Phylum.....Porifera (Porous body, cellular grade of body Organization, radially symmetrical).
Class.....Calcarea (Skeleton having calcareous spicules)
OrderHomocoela (Ascon type canal system)
Genus*Leucosolenia*

COMMENT:

1. It is a marine colonial sponge inhabiting shallow waters below tide mark, not found in calm water. It is abundant in Northern Atlantic Coast and very common in Europe.
2. The colony of *Leucosolenia* is whitish yellow in color consisting of vase like simple or branching bodies about 25 mm long, joined together by horizontal tubes at the base.
3. The body surface is perforated by numerous pores called Ostia or incurrent pores through this water enter the body of sponge.
4. Each vertical tube terminates into osculum at distal end through which water moves out of the body of sponge.
5. Calcareous spicules are monoaxon (spicules growing along single axis) or triaxon (spicules consisting of three axes crossing at right angles) type.
6. Simplest type of canal system i.e. Ascon type the course taken by water current is: Through Ostia through Osculum Ingressing water Spongocoel to outside
7. Asexual reproduction by Branching and Budding.
8. Sexual reproduction by ova and spermatozoa, indirect development including Paranchymula larva.

SYCON- SPONGE

SYSTEMATIC POSITION

Phylum.....Porifera (Porous body, cellular grade of body organization, radially symmetrical)

Class.....Calcarea (Skeleton having calcareous spicules)

OrderHeterocoela (Sycon type canal system)

COMMENTS:

1. Scypha is commonly called Crown sponge.
2. It is a small, solitary or colonial marine sponge found in shallow and well oxygenated water. Commonly distributed in Europe from Rhode Island to Greenland.
3. Body is vase-shaped (measuring 20-25 mm in length) opening to exterior by osculum at distal end.
4. Osculum is encircled by oculars fringe formed by monoaxon spicules.
5. Proximal end or base is attached to the substratum.
6. Body wall is thick having monoaxon, triaxon and tetraxon spicules.
7. Body wall consists of outer dermal epithelium and inner flattened epithelium which lines the spongocoel separated by a middle layer of mesenchyme.
8. Canal system is syconoid type. Water current passes through Ostia incurrent Canal Prosopyles radial canal apopyles spongocoel osculum exterior.
9. Reproduction by asexual (budding) as well as sexual method.
10. Larval stage is called amphiblastula.

AURELIA

SYSTEMATIC POSITION

Phylum ...Coelentrata (Tissue grade organization, diploplastic and acoelomate)

Class ... Scyphozoa (Medusa well developed, hydroid reduced)

Order ... Semaestomeae (Free swimming scyphomedusae with umbrella like disc and oral lobes)

Genus*Aurelia* (Jelly fish)

COMMENTS:

1. *Aurelia* is commonly known as Jelly-fish. It is a solitary, marine, medusoid form

Inhabiting warm and temperate seas throughout the world.

2. Body is transparent, bluish white in colour. Inner organs are visible from body surface.

3. Circular body measuring about 90 mm in diameter having convex aboral (exumbrellar) and concave oral (sub umbrella) surface.

4. Four red or purple horse-shoe shaped gonads are visible from aboral surface.

5. Mouth is four cornered each corner continues into four long and narrow oral lobes

Hanging down from oral surface.

6. Subumbrellar margin contain marginal tentacles having stinging cells and 8 marginal lappets having sense organs called tentaculocysts (rhopalium).

7. Jelly fish is carnivorous feeding on small animals like molluscs, crustaceans, protozoans, nematodes etc.

8. It swims by rhythmic contractions of umbrellar surface.

9. *Aurelia* is dioecious i.e. male and female individuals are separate.

10. Life cycle shows alternation of generation.

OBELIA

SYSTEMATIC POSITION

Phylum ...Coelentrata (Tissue grade organization, diploplastic and acoelomate)

Class ...Hydrozoa (Hydroids, medusa with velum)

Order ...Hydroidea (Polypoid generation well developed)

Genus ... *Obelia*

COMMENTS:

1. *Obelia* is commonly called as sea-fur. It is sedentary marine colonial form attached on the surface of sea weeds, molluscan shells, rocks and wooden piles in shallow waters. It is widely distributed throughout the world.
2. It is a trimorphic colony present in the form of filamentous sea weed measuring several centimeters in height.
3. The basal horizontal portion is known as hydrorhiza or stolon. It helps in attachment with the substratum. It gives up vertical branches called hydrocaulus.
4. Hydrorhiza and hydrocaulus are covered by chitinous perisarc which encloses soft inner coenosarc.
5. Coenosarc is the living, hollow cellular tube made up of ectoderm, endoderm and mesoglea.
6. The trimorphic colony of *Obelia* possess three types of zooids polyp (hydranths), gonangium (blastostyle) and medusa (sexual zooid).
7. Polyp is the nutritive zooid. It is a bell-shaped cup made up of lower hydrotheca and upper hypostome. Hypostome carries a circlet of numerous tentacles provided with nematocysts.
8. Blastostyle is the club shaped reproductive zooid enclosed within a covering called gonotheca. It gives rise to buds which develop into medusa.
9. Medusa is the free-swimming reproductive zooid consisting of upper ex-umbrella (Convex) and lowers sub-umbrella (concave) sides. It is provided with marginal Tentacles, four radial canals bearing gonads, a ring canal and a central hanging manubrium on the concave side.
10. It reproduces by asexual as well as sexual methods. Life history of *Obelia* exhibits alternation of generation.

PLANARIA

SYSTEMATIC POSITION

Kingdom: Animalia

Phylum: Platyhelminthes

Class: Rhabditophora

Clade: Adiaphanida

Order: Tricladida

Family: Planariidae

Genus: Planaria

COMMENTS:

1. Planaria is flatworms belonging to the phylum Platyhelminthes that have incredible regeneration ability, earning them the nickname "immortal under the knife."
2. Freshwater, marine, and terrestrial habitats all have a diverse range of organisms.
3. The majority of planarians lives in freshwater and can be found in huge groups; some species are marine, while others are terrestrial.
4. Some species are parasitic, meaning they feed on other living creatures' bodies.

FASCIOLA HEPATICA (Liver fluke)

SYSTEMATIC POSITION

Phylum ---Platyhelminthes - Flatworms, acoelomate

Class ---- Trematoda - Body wall without opidermis and cilia, are well developed.

Order ----Digenea - Endoparasitic,many larval stages present, Two suckers without hooks

Genus ----*Fasciola*

Species--- *hepatica*

COMMENTS:

1. *Fasciola hepatica* is found in the bile duct Body is leaf –like, dorso in breadth.
2. Anterior end is produced into a conical projection is known as cephalic cone.
3. Posterior end is large more rounded in front than behind.
4. Mouth is situated ventrally at anterior end and s
5. Digestive system comprised of pharynx
6. They are Hermaphrodite.
7. Excretory pore is found at the posterior extremity.
8. Eggs pass to the exterior through median genital pore situated between ora Sucker.
9. Life Cycle comprises of an intermediate host
10. They cause a disease known as liver rot.

TAENIA SOLIUM

SYSTEMATIC POSITION

Phylum: Platyhelminthes: Flatworms, acoelomate

Class - Cestoda

Subclass - EuCestode: Ribbon like, anterior end with an expanded scolex having

Order - Taenioidea: Endoparasite, scolex with four

Genus- *Taenia*

Species- *solium*

COMMENTS:

1. Each Mature Proglottid is square in shape.
2. Lateral side of each Proglottid contains lateral longitudinal nerve cords and lateral excretory canals.
3. A set of male and female reproductive organs is present in each mature Proglottid.
4. Male genital system consists of follicular testes, vasa efferentia, vas deferens and cirrus.
5. Female genital system consists of ovarian lobes (bilobed ovary), oviduct, ootype, vagina, uterus, Mehli's glands and compact vitellaria.
6. Oviducts divide into two ducts, one leading to vagina opening and other into uterus.
7. Vitelline gland lies behind ovary.
8. Male and female reproductive organs open into genital atrium.
9. The genital atrium opens by a common gonopore on a swollen genital papilla.

COCKROACH

SYSTEMATIC POSITION

Kingdom- Animalia

Phylum- Arthropoda

Class- Insecta

Order- Blattodea

Family- Blattidae

Genus- Periplaneta

Species- Americana

COMMENTS:

- 1) Cockroaches are scientifically known as *Periplaneta americana*.
- 2) They belong to the kingdom group of invertebrates.
- 3) They have articulating appendages that place them in the phylum Arthropoda, the largest phylum in the animal kingdom.
- 4) Having six pairs of legs and a pair of wings allows taxonomists to assign them to the Insecta class.
- 5) These creatures may be found all throughout the world and enjoy wet, dark, and moist environments.

PHERETIMA (EARTH WORM)

SYSTEMATIC POSITION:

Phylum Annelida
Class Oligochaeta
Order Neoligochaeta
Genus *Pheretima*

COMMENTS:

1. *Pheretima* is commonly called earthworm.
2. Body is bilaterally symmetrical, narrow, long, elongated and cylindrical measuring up to 150 mm in length.
3. Anterior end is tapering and posterior end is blunt.
4. Body is divided into 100-120 ring-like segments by a distinct series of annular grooves.
5. Each segment is provided with setae arranged in a ring with each seta arising from a setigerous sac of the skin. The setae help in locomotion by holding the earth.
6. Mouth is crescentic aperture situated just below the prostomium.
7. The clitellum is a circular band of glandular tissue which completely surrounds the segments from 14th to 16th segment.
8. Hermaphrodite. A pair of male genital pore is situated ventrally in the eighteenth segment while female genital pore is situated at the ventral surface of fourteenth segment.
9. Anus is situated at the terminal end of the last body segment called anal segment.
10. Reproduction is usually sexual.
11. Development takes place in cocoons.
12. Earthworm is used as bait in fishing and as food by many uncivilized people. It also has use in medicines, education, experiments and in agriculture as producer of organic fertilizer.

NEREIS

SYSTEMATIC POSITION:

Kingdom:	Animalia
Phylum:	Annelida
Subclass:	Errantia
Order:	Phyllodocida
Family:	Nereididae
Genus:	<i>Nereis</i>

COMMENTS:

- 1) *Nereis* is a genus of polychaete worms in the family Nereididae.
- 2) It comprises many species, most of which are marine.
- 3) *Nereis* possesses setae and parapodia for locomotion and gas exchange.^[1] They may have two types of setae, which are found on the parapodia.
- 4) Acicular setae provide support. Locomotors setae are for crawling, and are the bristles that are visible on the exterior of the Polychaeta.
- 5) They are cylindrical in shape, found not only in sandy areas, and they are adapted to burrow.
- 6) They often cling to sea grass (posidonia) or other grass on rocks and sometimes gather in large groups.
- 7) Nereis worms are commonly known as rag worms or clam worms.
- 8) The body is long, slender, and dorso-ventrally flattened, reaching a length of 5 - 30 cm.
- 9) The head consists of two parts: a roughly triangular anterior lobe the prostomium and a posterior ring-like portion—the peristomium.
- 10) The latter bears four pairs of tentacular cirri, dorsally two pairs of eyes, and ventrally a pair of short two-jointed palps.

HIRUDINARIA

SYSTEMATIC POSITION:

Phylum	Annelida
Class	Hirudinea
Order	Gnathobdellida
Genus	<i>Hirudinaria</i>

COMMENTS:

1. *Hirudinaria granulosa* is commonly called Indian cattle leech.
2. Body is soft, elongated, vermiform and bilaterally symmetrical with dorsal surface green and ventral surface orange yellow in colour.
3. During extension of the body is dorso-ventrally flattened while during contraction it is cylindrical.
4. Body of leech is divided into 33 segments. The segments are further divided into annuli or rings.
5. Anterior and posterior suckers are well developed. Anterior sucker is oval bearing triradiate mouth and is formed by the fusion of prostomium with a few somites of anterior region.
6. The posterior sucker is circular in outline and forms a muscular disc at the posterior end. Both the suckers serve as powerful organs of adhesion and locomotion.
7. Five pairs of eyes are present on the dorsal side.
8. No special respiratory organs. Respiration happens via skin.
9. Locomotion by creeping/crawling on the surface.
10. Alimentary canal is a straight tube extending from mouth to anus. Small aperture or anus is situated mid dorsally on the 26th segment at the root of the posterior sucker.
11. Excretory system consists of segmentally arranged seventeen pairs of nephridia arranged from six to twenty-two segments, opening by nephridiopores on the ventral surface.
12. Hermaphrodite. Male genital aperture is situated mid-ventrally in the second and third annuli of 10th segment and female genital aperture between second and third annuli of 11th segment.
13. Sexual reproduction is common.

PALAEEMON (PRAWN)

SYSTEMATIC POSITION:

Phylum Arthropoda

Class Crustacea

Order Decapoda

Genus *Palaemon*

COMMENTS:

1. *Palaemon* is commonly known as prawn.
2. Body is elongated, spindle shaped and bilaterally symmetrical.
3. *Palaemon* species are of pale-yellow, pale-blue and greenish color with brown tinge or with orange-red patches. Preserved specimens become deep orange-red.
4. The body can be divided into two regions, anterior cephalothorax and posterior abdomen.
5. Cephalothorax is a large, rigid, unjointed, immovable and cylindrical structure. It consists of 13 segments, 5 of the head regions, and 8 of the thorax regions (Kotpal, 2005).
6. Abdomen is rounded, jointed and compressed laterally. It consists of 6 movable segments and a terminal conical structure, called telson. Each abdominal segment bears a pair of jointed appendages called pleopods or swimmerets.
7. The prawn uses its walking legs for movement at the water-bed.
8. Respiration happens *via* gills, epipodites and lining of branchiostegites.
9. Excretion through a pair of antennary or green glands, a pair of lateral ducts, an unpaired renal or nephroperitoneal sac and the integument.
10. Sexes are separate. Sexual dimorphism is well developed

PLANARIA

SYSTEMATIC POSITION:

Phylum: Platyhelminthes

Domain: Eukaryota

Kingdom: Animalia

Order: Tricladida; Lang, 1884

Subphylum: Rhabditophora

COMMENTS

1. A **planarian** is one of the many flatworms of the traditional class Turbellaria.
2. It usually describes free-living flatworms of the order **Tricladida (triclads)**, although this common name is also used for a wide number of free-living Platyhelminthes.
3. Planaria are common to many parts of the world, living in both saltwater and freshwater ponds and rivers. Some species are terrestrial and are found under logs, in or on the soil, and on plants in humid areas.
4. Planaria exhibit an extraordinary ability to regenerate lost body parts. For example, a planarian split lengthwise or crosswise will regenerate into two separate individuals.
5. Some planarian species have two eye-spots (also known as ocelli) that can detect the intensity of light, while others have several eye-spots.
6. The eye-spots act as photoreceptors and are used to move away from light sources. Planaria have three germ layers (ectoderm, mesoderm, and endoderm), and are acoelomate (they have a very solid body with no body cavity).
7. They have a single-opening digestive tract; in Tricladida planarians this consists of one anterior branch and two posterior branches.

PALAMNAEUS (SCORPION)

SYSTEMATIC POSITION:

Phylum Arthropoda

Class Arachnida

Order Scorpionidea

Genus *Palamnaeus*

COMMENTS:

1. *Palamnaeus* is commonly known as scorpion.
2. Body is bilaterally symmetrical, triploblastic, coelomate, metamerically segmented animal with jointed appendages.
3. Body is differentiated into anterior prosoma, middle mesosoma and posterior metasoma.
4. Prosoma is covered by carapace that bears a pair of median eyes and 2-5 pairs of lateral eyes.
5. It has six segments, each with a pair of appendages, *i.e.*, one pair of chelicerae, one pair of pedipalp and four pairs of walking legs.
6. Opisthosoma consists of two parts anterior mesosoma and posterior metasoma.
7. Mesosoma is seven segmented, the first segment bears genital operculum, the second segment bears a pair of pectin's and each of the third, fourth, and fifth and sixth segments have a pair of stigmata. The seventh segment is without any appendages.
7. Metasoma is tail like and comprises five caudal segments and a last segment or telson bearing the sting.
8. Sexes are separate. Viviparous.
9. *Palamnaeus* are harmful to mankind; its sting can cause swelling at the sting site, extreme pain, fever and in some cases collapse.

GRASS HOPPER

SYSTEMATIC POSITION

Domain:	Eukaryota
Kingdom:	Animalia
Phylum:	Arthropoda
Class:	Insecta
Order:	Orthoptera

COMMENTS:

- 1) Grasshoppers are a group of insects belonging to the suborder Caelifera.
- 2) They are among what is possibly the most ancient living group of chewing herbivorous insects, dating back to the early Triassic around 250 million years ago.
- 3) Grasshoppers are typically ground-dwelling insects with powerful hind legs which allow them to escape from threats by leaping vigorously.
- 4) Their front leg is shorter and used for grasping food. As hemimetabolous insects, they do not undergo complete metamorphosis; they hatch from an egg into a nymph or "hopper" which undergoes five moults, becoming more similar to the adult insect at each developmental stage.
- 5) The grasshopper hears through the tympanal organ which can be found in the first segment of the abdomen attached to the thorax; while its sense of vision is in the compound eyes, the change in light intensity is perceived in the simple eyes (ocelli).
- 6) At high population densities and under certain environmental conditions, some grasshopper species can change color and behavior and form swarms. Under these circumstances, they are known as locusts.

LAMELLIDENS

SYSTEMATIC POSITION:

Phylum : Mollusca

Class : Pelecypoda (Bivalvia)

Order : Eulamellibranchiata

Genus : Lamellidens

Species : marginalis

PHYLUM: MOLLUSCA

1. Mollusca are soft bodied animals.
2. They are triploblastic animals.
3. The mollusks are provided with one or two calcareous shells.
4. The shells may be external or internal.
5. They have bilateral symmetry.
6. The shell of two valves and hence the name Bi valvia.
7. Inter filamentary junctions and inter lamellar junctions are vascular.
8. The gills are basket- like.

PILA GLOBOSA

SYSTEMATIC POSITION

Phylum Mollusca

Class Gastropoda

Order Mesogastropoda

Genus *Pila*

Species *globosa*

COMMENTS:

1. *Pila globosa* is commonly called apple snail.
2. Shell is univalve and coiled around a central axis in a right-handed spiral.
3. Operculum is well developed and closes the aperture or the mouth of the shell.
4. The body is soft and is enclosed in a shell. It consists of head, foot and visceral mass.
5. Head is the anterior fleshy part of the body overhanging the foot. It bears mouth, two pairs of contractile tentacles and a pair of eyes.
6. Foot is the locomotory organ of *Pila* and lies below the head.
7. All visceral organs are contained in this lump like structure that lies above the head-foot complex.
8. Skin of the visceral mass forms a thin and delicate covering called the mantle.
9. Respiratory organs consist of a single ctenidium or gill, a pulmonary sac or lung and a pair of nuchal lobes. Aquatic respiration by ctenidium and aerial respiration by pulmonary chamber.
10. Sexes are separate but without sexual dimorphism.
11. It is found in large numbers in those areas which are rich in aquatic plants like *Vallisneria* and *Pistia*

ASTERIAS (STAR FISH)

SYSTEMATIC POSITION:

Phylum: Echinodermata

Class : Asteroidea

Order : Forcipulata

Genus : *Asterias*

Species: *rubens*

COMMENTS:

1. *Asterias* is commonly known as starfish or sea star
2. Average size of *Asterias* ranges from 10 to 25 cm in diameter.
3. The body is star shaped, consisting of a central disc with five radiating arms which are broad at their base and tapering towards their extremities.
4. Mouth is pentagonal and lies in the center of the disc on oral surface.
5. Aboral surface bears a number of short, stout spines arranged in irregular rows, numerous dermal
6. Branchia among spines and a smaller aperture, the anus which is situated near the center.
7. Pedicellariae are very small, microscopic bodies scattered all over the body.
8. Water vascular system is well developed.
9. Sexes are separate. Fertilization is external.
10. Development includes a free swimming bippinnaria larva.

BRANCHIOSTOMA (AMPHIOXUS)

SYSTEMATIC POSITION

Kingdom: Animalia

Phylum: Chordata

Subphylum: Cephalochordata

Class: Leptocardii

Order: Amphioxiformes

COMMENTS:

1. These five synapomorphies include a notochord, dorsal hollow nerve cord, endostyle or thyroid, pharyngeal slits, and a post-anal tail.
2. Lancelets contain many organs and organ systems that are closely related to those of modern fish, but in more primitive form. Therefore, they provide a number of examples of possible evolution exaptation.
3. The lancelets, also known as amphioxi (singular, amphioxus), consist of about 30-35 species of "fish-like" benthic filter feeding chordates in the order Amphioxiformes.

SHARK

SYSTEMATIC POSITION

Kingdom:	Animalia
Phylum:	Chordata
Class:	Chondrichthyes
Infraclass:	Euselachii
Clade:	Neoselachii
Subdivision:	Selachimorp

COMMENTS:

1. Sharks range in size from the small dwarf lantern shark (*Etmopterus perryi*), a deep sea species that is only 17 centimeters (6.7 in) in length, to the whale shark (*Rhincodon typus*), the largest fish in the world, which reaches approximately 12 meters (40 ft) in length.
2. They are found in all seas and are common to depths up to 2,000 meters (6,600 ft).
3. They generally do not live in freshwater, although there are a few known exceptions, such as the bull shark and the river shark, which can be found in both seawater and freshwater.
4. Sharks have a covering of dermal denticles that protects their skin from damage and parasites in addition to improving their fluid dynamics.
5. They have numerous sets of replaceable teeth.

CATLA

SYSTEMATIC POSITION

Kingdom:	Animalia
Phylum:	Chordata
Class:	Actinopterygii
Order:	Cypriniformes
Family:	Cyprinidae
Subfamily:	Labeoninae
Genus:	Labeo
Species:	<i>L. catla</i>

COMMENTS:

1. Catla fish have distinctive looks with large and broad head.
2. They have a large protruding lower jaw, and upturned mouth.
3. Scales of these fish are large and they have greyish scales on their dorsal side.
4. The scales are whitish on their belly. Their body is short and deep, somewhat laterally compressed.
5. Their eyes are large and visible from underside of the head.
6. Lower lip of the Catla fish is very thick, and the upper lip is absent. Their anal fin is short, dorsal fin inserted slightly in advance of pelvic fins.
7. The caudal fin is forked, and the pectoral fins are long extending to pelvic fins. Fish with average body weight of 1-2 kg is generally sold in the market. But Catla fish can grow up to 20 kg.
8. Their body length can reach around 1 meter in aged and large sized fish

RANA HEXADACTYLA

SYSTEMATIC POSITION

Phylum	: Chordata
Class	: Amphibia
Family	: Ranidae
Order	: Anura
Genus	: <i>Rana</i>
Species	: <i>hexadactyla</i>
Phylum	: Chordata
Class	: Amphibia

COMMENTS:

1. They contain a cranium around the brain.
2. Notochord is replaced by vertebral column.
3. Pharyngeal gill slit.
4. It is cold blooded vertebrates with dual life
5. They have jaws.
6. The limbs are pentadactylous.
7. Body short, broad and trunk fused.
8. Neck and tail absent
9. Fore limbs short, hind limbs long and help in long jump.

SALAMENTOR

SYSTEMATIC POSITION

Domain : Eukaryota
Kingdom: Animalia
Phylum : Chordata
Class : Amphibia
Clade : Caudata

COMMENTS:

1. Urodela **Salamanders** are a group of amphibians typically characterized by their lizard-like appearance, with slender bodies, blunt snouts, short limbs projecting at right angles to the body, and the presence of a tail in both larvae and adults.
2. All ten extant salamander families are grouped together under the order Urodela from the group Caudata.
3. Salamanders rarely have more than four toes on their front legs and five on their rear legs, but some species have fewer digits and others lack hind limbs.
4. Their permeable skin usually makes them reliant on habitats in or near water or other cool, damp places.
5. This group of amphibians is capable of regenerating lost limbs as well as other damaged parts of their bodies.
6. Researchers hope to reverse engineer the Regenerative processes for potential human medical applications, such as brain and spinal cord injury treatment or preventing harmful scarring during heart surgery recovery.

CALOTES VERSICOLAR

SYSTEMATIC POSITION

Phylum : Chordata
Class : Reptilia
Order : Squamata
Genus : *Calotes*
Species : *versicolor*

COMMENTS:

1. Notochord is replaced by vertebral column
2. Distinct cranium or brain case is present.
3. Presence of well-developed head.
4. Skin dry, cornified, usually covered with epidermal scales or scutes.
5. A post anal tail is usually present.
6. Well-developed eyes, provide with eyelids and nictitating membrane.
7. Body is covered by epidermal scales
8. Teeth are pleurodont.
9. Quadrate is movable.
10. Cloacal opening is transverse

CHAMELEON

SYSTEMATIC POSITION

Kingdom: Animalia
Phylum: Chordata
Class: Reptilia
Order: Squamata
Suborder: Iguania
Clade: Acrodonta
Family: Chamaeleonidae

COMMENTS:

1. The chameleon spirit animal has a shining personality, always displaying courage and boldness with every human contact.
2. The chameleon meaning also speaks about having a clearer perception and a stronger intuition.
3. The chameleon symbolism represents a period of changes. You need to learn to adjust to your environment and blend in to create harmony.
4. Avoid unnecessary conflicts and confrontations. Do not rock the boat but foster an environment where you can feel secure and stable.
5. Unlike the manatee spirit animal, the chameleon meaning is about being focused. Keep your eyes on the prize and your actions in check.
6. Focusing on your higher ambitions and goals. Pay attention to what keeps your feet on the ground.
7. The chameleon spirit animal symbolizes sensitivity and clairvoyance. It encourages you to have patience, to gain insight, to always be curious, and to be resourceful.
8. It reminds you that there is no need to go through life in a hurry because you should live it at your own pace.

TURTLE

SYSTEMATIC POSITION

Kingdom: Animalia
Phylum: Chordata
Class: Reptilia
Clade: Pantestudines
Clade: Testudinata
Clade: Perichelydia
Order: Testudines

COMMENTS:

1. Turtles are an order of reptiles known as Testudines, characterized by a special shell developed mainly from their ribs.
2. Modern turtles are divided into two major groups, the Pleurodira (side necked turtles) and Cryptodira (hidden necked turtles), which differ in the way the head retracts.
3. There are 360 living and recently extinct species of turtles, including land-dwelling tortoises and freshwater terrapins.
4. They are found on most continents, some islands and, in the case of sea turtles, much of the ocean. Like other amniotes (reptiles, birds, and mammals) they breathe air and do not lay eggs underwater, although many species live in or around water.
5. Turtle shells are made mostly of bone; the upper part is the domed carapace, while the underside is the flatter plastron or belly-plate.
6. Its outer surface is covered in scales made of keratin, the material of hair, horns, and claws.
7. The carapace bones develop from ribs that grow sideways and develop into broad flat plates that join up to cover the body.
8. Turtles are ectotherms or "cold-blooded", meaning that their internal temperature varies with their direct environment.
9. They are generally opportunistic omnivores and mainly feed on plants and animals with limited movements.
10. Many turtles migrate short distances seasonally. Sea turtles are the only reptiles that migrate long distances to lay their eggs on a favored beach.

COBRA

SYSTEMATIC POSITION

Kingdom: Animalia

Phylum: Chordata

Class: Reptilia

Order: Squamata

Suborder: Serpentes

Family: Elapidae

COMMENTS:

1. Naia is most common and deadly or poisonous snake of India. It lives in burrows, understones, in thick vegetation and may also come in human dwellings.
2. It grows upto 6-8 meters.
3. It is brown or brownish black. It has small and indistrict head. The neck can dilate (due to raising of cervical ribs) and form a hood. It may or may not bear spectacle marks dorsally.
4. The pupil around, the 3rd supralabial shield of upper lip touches eye and nasal shield. The subcaudal shields are arranged in 2 rows.
5. The fangs are anterior, grooved and permanently erect.
6. When annoyed, it raises the front part of the body, spreads its hoods and hisses through the nose. It bites with a chewing motion making several wounds.
7. The venom is neurotoxic and fatal being more deadly than that of vipers

VIPER

SYSTEMATIC POSITION

Kingdom: Animalia
Phylum: Chordata
Class: Reptilia
Order: Squamata
Suborder: Serpentes
Clade: Colubroides
Family: Viperidae

COMMENTS

1. The Viperidae (vipers) are a family of snakes found in most parts of the world, except for Antarctica, Australia, Hawaii, Madagascar, New Zealand, and various other isolated islands.
2. They are venomous and have long (relative to non-vipers), hinged fangs that permit deep penetration and injection of their venom.
3. Three subfamilies are currently recognized.
4. They are also known as viperids.
5. The name "viper" is derived from the Latin word *vipera*, -ae, also meaning viper, possibly from *vivus* ("living") and *parere* ("to beget"), referring to the trait viviparity (giving live birth) common in vipers like most of the species of Boidae.

PIGION (*Columba liva*)

SYSTEMATIC POSITION:

Phylum : Chordata
Class : Aves
Order : Columbiformes
Genus : *Columba*
Species : *liva*

COMMENTS

1. Notochord is replaced by vertebral column
2. Distinct cranium or brain case is present.
3. Presence of well-developed head.
4. Body is covered with Feathers
5. Head is small, fairly long movable neck and mouth is provided with beaks.
6. Head and neck are small with short, slender beak.
7. Skin soft and thick, wings long and pointed.

RAT

SYSTEMATIC POSITION:

Kingdom: Animalia
Phylum: Chordata
Class: Mammalia
Clade: Simplicidentata
Order: Rodentia

COMMENTS:

1. Commonly called as black rat also known as ship rat, roof rat, or house rat.
2. Body divisible into head, neck, trunk and tail.
3. Head contains ears, eyes, nostrils, and snout with long moustache or vibrissae.
4. Body and limbs covered with hairs.
5. Pinnae well developed. Eyes sharp.
6. This black rat exhibits several color forms. It is usually black to light brown in color with a lighter underside.
7. Tail is elongated and scaly.
8. Incisor teeth chisel-like, open-rooted, used for gnawing. Canines absent.
9. It is a predatory species and adapts to different micro-habitats.
10. Black rats (or their ectoparasites) can carry a number of pathogens, of which bubonic plague (via the Oriental rat flea), typhus, Weil's disease, toxoplasmosis and trichinosis are the best known.
11. They are great climbers and prefer to live in palms and trees, such as pine trees.
12. Viviparous.

BAT

SYSTEMATIC POSITION

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
<i>Clade:</i>	Scrotifera
Order:	Chiroptera

COMMENTS

1. Conservation is important not only for biodiversity but also because these flying mammals provide ecological services essential for humans.
2. Bats are very useful for the control of insect populations and specifically pests to agriculture.
3. Their diet very often includes Lepidoptera, a large order with several species very harmful to many plants of great economic interest.
4. Insect-eating bats are great for keeping bugs away from crops, as well as the places where the bats roost. The Brazilian free-tailed bat has been recognized as an important "pest management service" in cotton farming. Because bats eat so many insects in some regions, they can also reduce the need for pesticide sprays.
5. Natural enemies of insect pests, also known as biological control agents, include predators, parasitoids, pathogens, and competitors.
6. Bats take the night shift, playing a major role in pollinating crops and spreading seeds.
7. Fruit-eating bats play important roles in distributing seeds to maintain plants and forests.
8. These species of bats, often called "flying foxes" because of their larger body size and big eyes, live in tropical and subtropical areas of the Old World (Africa, Asia and Australia).
9. Bats are of major environmental significance due to being important indicators of how well an ecosystem is doing.
10. They are warm-blooded, suckle their young, and only have one pup a year, which means the population growth, can decline dramatically if weather is particularly harsh and the pups are not able to survive.

RABBIT

SYSTEMATIC POSITION

Phylum :- chordata (notochord, dorsal tubular nerve cord present and gill slits present)

Group :- craniata (definite head. cranium with brain present)

Sub-phylum :- vertebrata (vertebral column present)

Division :- gnathostomata (jaws and paired appendages present)

Super class :- tetrapoda (paired limbs, lungs, bony skeleton and cornified skin).

Class :-mammalia (body covered with hairs. females have mammary glands).

Sub-class :- theria (viviparous).

Infraclass:- eutheria (placental mammals, vagina single).

Order:- lagomorpha (7 upper incisors 2 pairs).

Genus: *oryctolagus*

COMMENTS

1. Commonly called as Rabbit.
2. Body cat-like and divisible into head, neck, trunk abdomen and tail.
3. Head contains long tactile vibrissae or whiskers, external nares, usually shorter eyes and mouth.
4. External ears large having external auditory meatus.
5. The long ears of rabbits are most likely an adaptation for detecting predators.
6. Length 40 cm from mouth to anus.
7. Fur colour white, black, brown or spotted.
8. Males have muscular skin-covered penis. Females have clitoris.
9. Forelimbs used for digging and hind limbs for leaping. Fast runner (30 to 40 km per hour). Forelimbs contain upper arm, forearm, metacarpus, hand and clawed fingers. Hind limb contains thigh, shank, metatarsus, foot and clawed toes.
10. Active throughout the year; no species is known to hibernate. Rabbits are generally nocturnal, and they also are relatively silent.
11. Instead of sound, scent seems to play a predominant role in the communication systems of most rabbits; they possess well-developed glands throughout their body and rub them on fixed objects to convey group identity, sex, age, social and reproductive status, and territory ownership.

COLOR BLINDNESS

1. Color blindness or color vision deficiency (CVD) is the decreased ability to see color or differences in color.
2. It can impair tasks such as selecting ripe fruit, choosing clothing, and reading traffic lights.
3. Color blindness may make some academic activities more difficult. However, issues are generally minor, and people with colorblindness automatically develop adaptations and coping mechanisms.
4. People with total color blindness (achromatopsia) may also be uncomfortable in bright environments and have decreased visual acuity.
5. The most common cause of color blindness is an inherited problem or variation in the functionality of one or more of the three classes of cone cells in the retina, which mediate color vision.
6. The most common form is caused by a genetic disorder called congenital red–green color blindness.
7. Males are more likely to be color blind than females, because the genes responsible for the most common forms of color blindness are on the X chromosome.
8. Females who are not color-blind can carry genes for color blindness and pass them on to their children.
9. Color blindness can also result from physical or chemical damage to the eye, the optic nerve, or parts of the brain.
10. Screening for color blindness is typically done with the Ishihara color test.

DOWN'S SYNDROME

1. Down syndrome or Down's syndrome, also known as trisomy 21, is a genetic disorder caused by the presence of all or part of a third copy of chromosome 21.
2. It is usually associated with developmental delays, mild to moderate intellectual disability, and characteristic physical features.
3. There are three types of Down syndrome, all with the same features: Trisomy 21, the most common type; Mosaic Down syndrome, and Translocation Down syndrome.
4. The parents of the affected individual are usually genetically normal. The probability increases from less than 0.1% in 20-year-old mothers to 3% in those of age 45.
5. The extra chromosome is provided at conception as the egg and sperm combine. A very small percentage of 1 to 2% gets the additional chromosome in the embryo stage and it only impacts some of the cells in the body; this is known as Mosaic Down syndrome.
6. Usually, babies get 23 chromosomes from each parent for a total of 46, whereas in Down syndrome, a third 21st chromosome is attached.
7. It is believed to occur by chance, with no known behavioral activity or environmental factor that changes the probability.
8. Down syndrome can be identified during pregnancy by prenatal screening, followed by diagnostic testing, or after birth by direct observation and genetic testing.
9. Since the introduction of screening, Down syndrome pregnancies are often aborted (rates varying from 50% to 85% depending on maternal age, gestational age, and maternal race/ethnicity)

HAEMOPHILIA

1. Hemophilia (from is a mostly inherited genetic disorder that impairs the body's ability to make blood clots, a process needed to stop bleeding
2. This results in people bleeding for a longer time after an injury, easy bruising, and an increased risk of bleeding inside joints or the brain.
3. Those with a mild case of the disease may have symptoms only after an accident or during surgery.
4. Bleeding into a joint can result in permanent damage while bleeding in the brain can result in long term headaches, seizures, or a decreased level of consciousness.
5. There are two main types of hemophilia: hemophilia, which occurs due to low amounts of clotting factor VIII, and hemophilia B, which occurs due to low levels of clotting factor IX.
6. They are typically inherited from one's parents through an X chromosome carrying a nonfunctional gene.
7. Rarely a new mutation may occur during early development or hemophilia may develop later in life due to antibodies forming against a clotting factor.
8. Other types include hemophilia C, which occurs due to low levels of factor XI, Von Willebrand disease, which occurs due to low levels of a substance called von Willebrand factor, and para hemophilia, which occurs due to low levels of factor V.
9. Hemophilia A, B, and C prevent the intrinsic pathway from functioning properly; this clotting pathway is necessary when there is damage to the endothelium of a blood vessel.
10. Acquired hemophilia is associated with cancers, autoimmune disorders, and pregnancy.

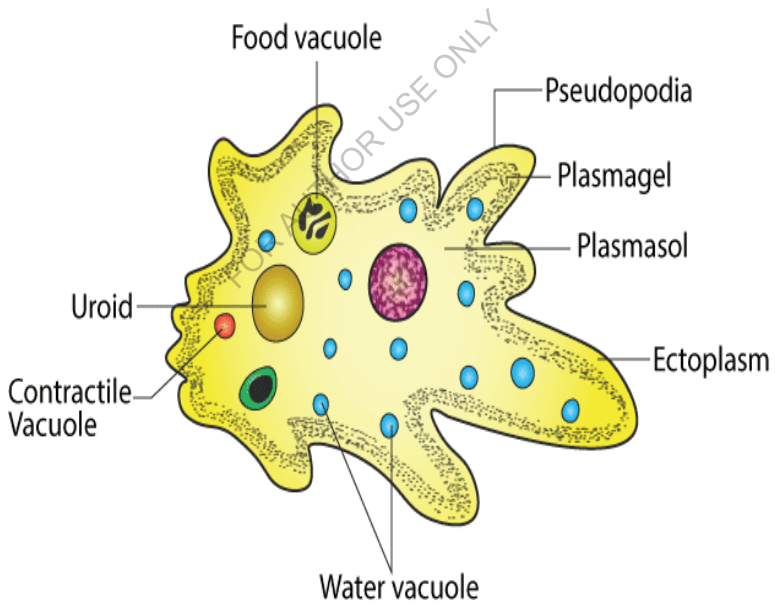
KLINEFELTERS SYNDROME

1. Klinefelter syndrome (ks), also known as 47, xxy, is an aneuploid genetic condition where a male has an additional copy of the x chromosome.
2. The primary features are infertility and small, poorly functioning testicles. usually, symptoms are subtle and subjects do not realize they are affected.
3. Sometimes, symptoms are more evident and may include weaker muscles, greater height, poor motor coordination, less body hair, breast growth, and less interest in sex.
4. These symptoms are noticed only at puberty. Intelligence is usually average, but reading difficulties and problems with speech are more common.
5. Klinefelter syndrome occurs randomly the extra x chromosome comes from the father and mother nearly equally.
6. An older mother may have a slightly increased risk of a child with the syndrome is defined by the presence of at least one extra x chromosome in addition to a y chromosome yielding a total of 47 or more chromosomes rather than the usual 46.
7. ks is diagnosed by the genetic test known as a karyotype.

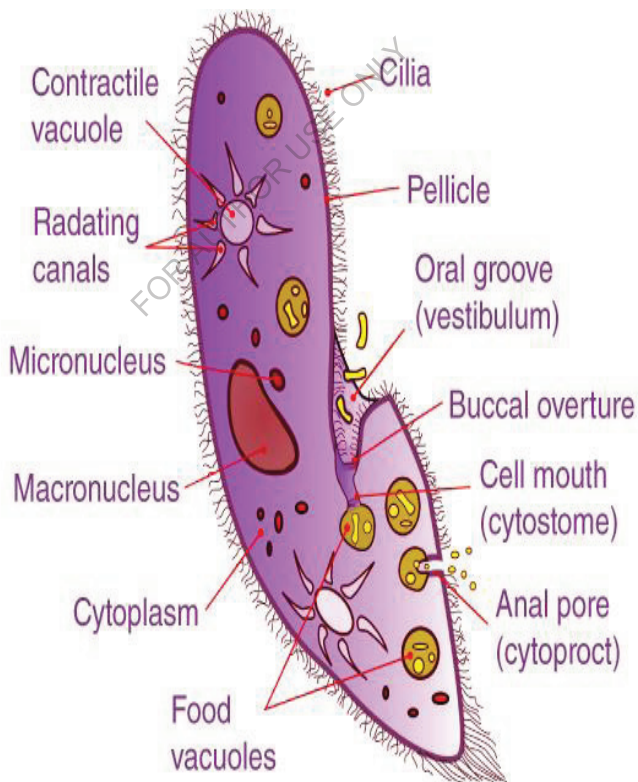
FOR AUTHOR USE ONLY

DIAGRAMS

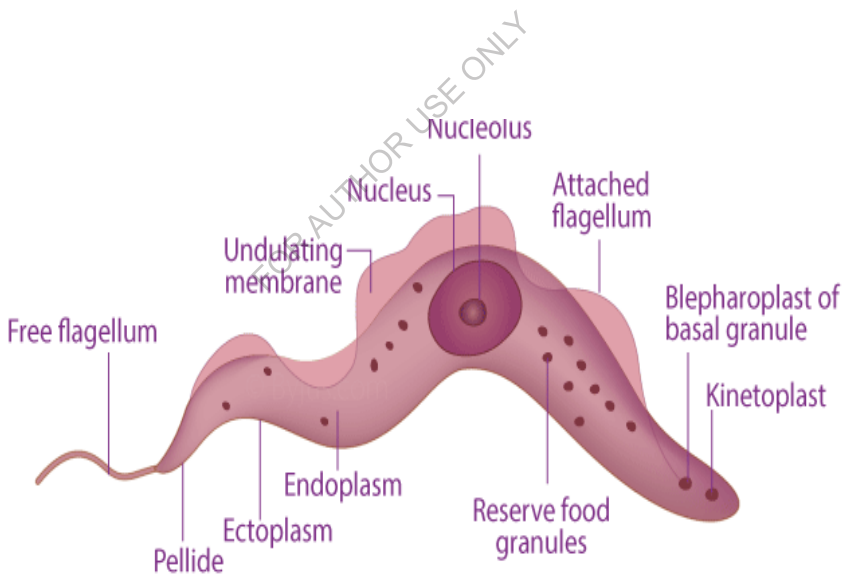
AMOEBA



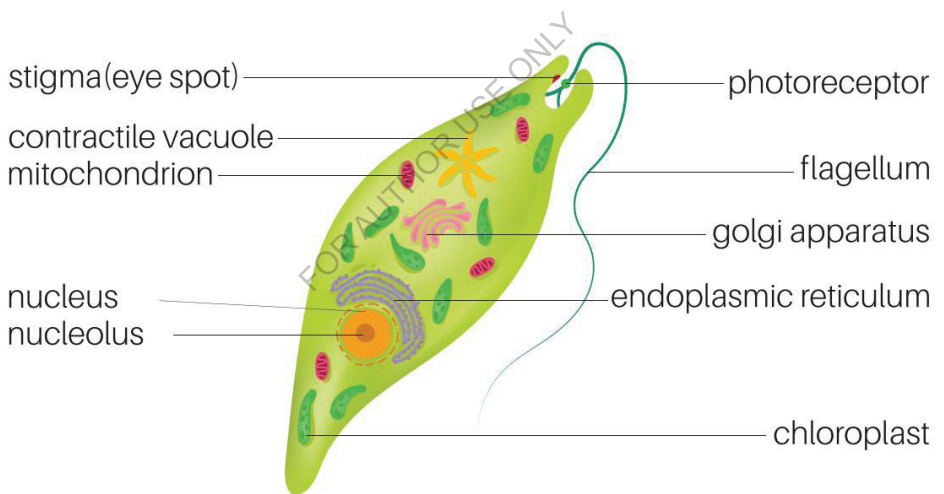
PARAMOECIUM



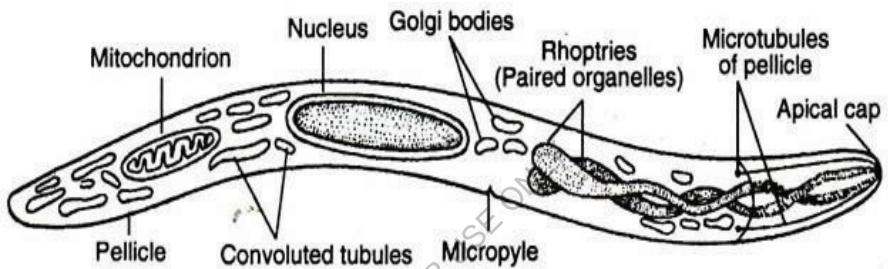
TRYPANOSOMA



EUGLENA

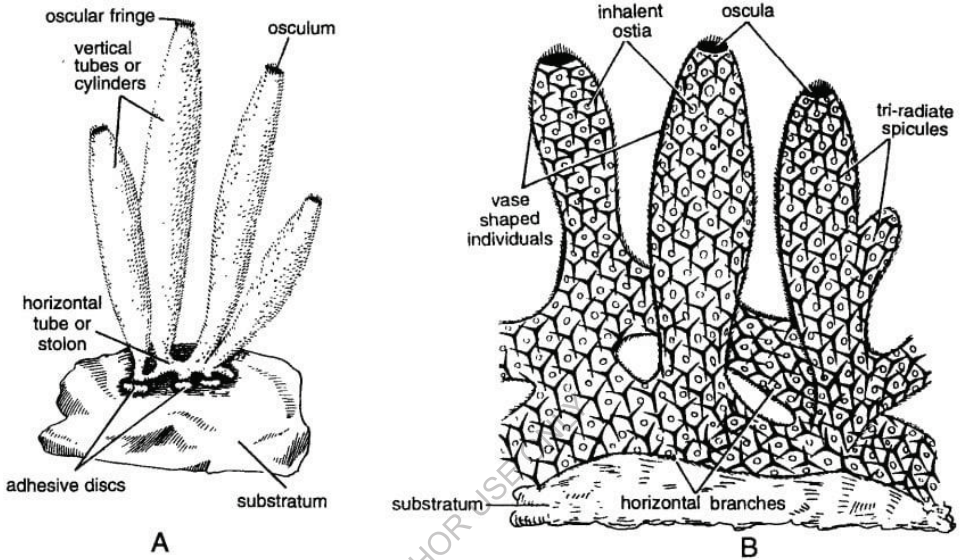


PLASMODIUM



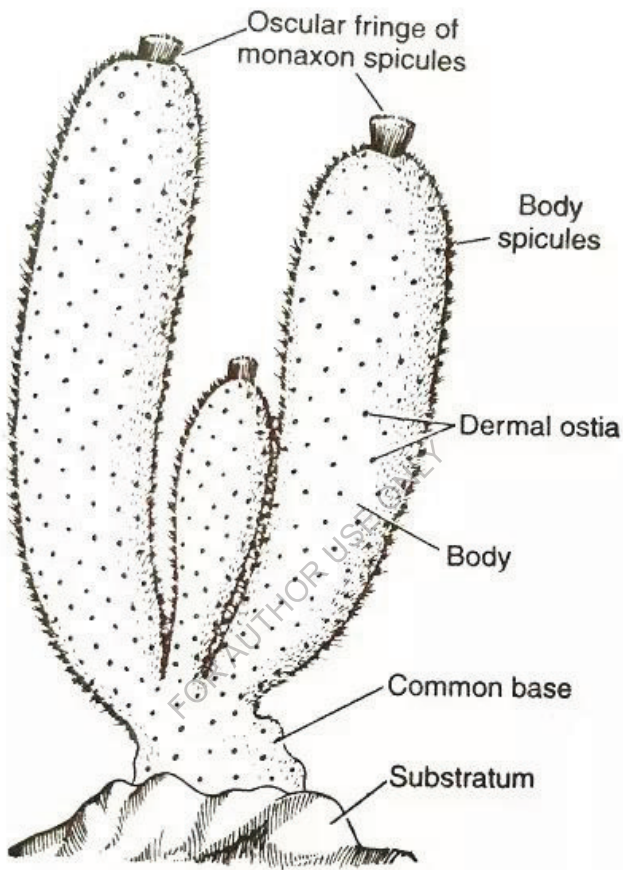
Ultrastructure of sporozoite of *Plasmodium*, seen in electron microscope

LEUCOSOLENIA

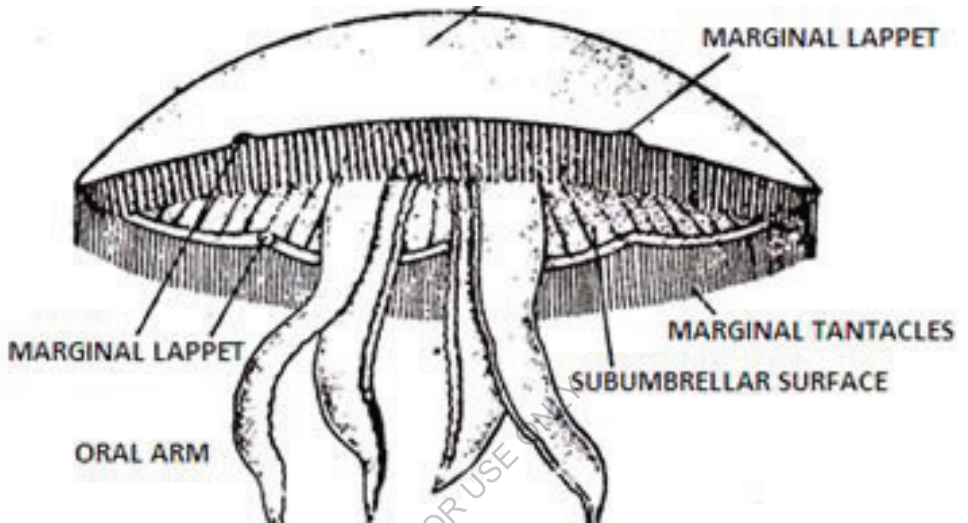


Leucosolenia. A. Colony, B. Portion of colony (magnified).

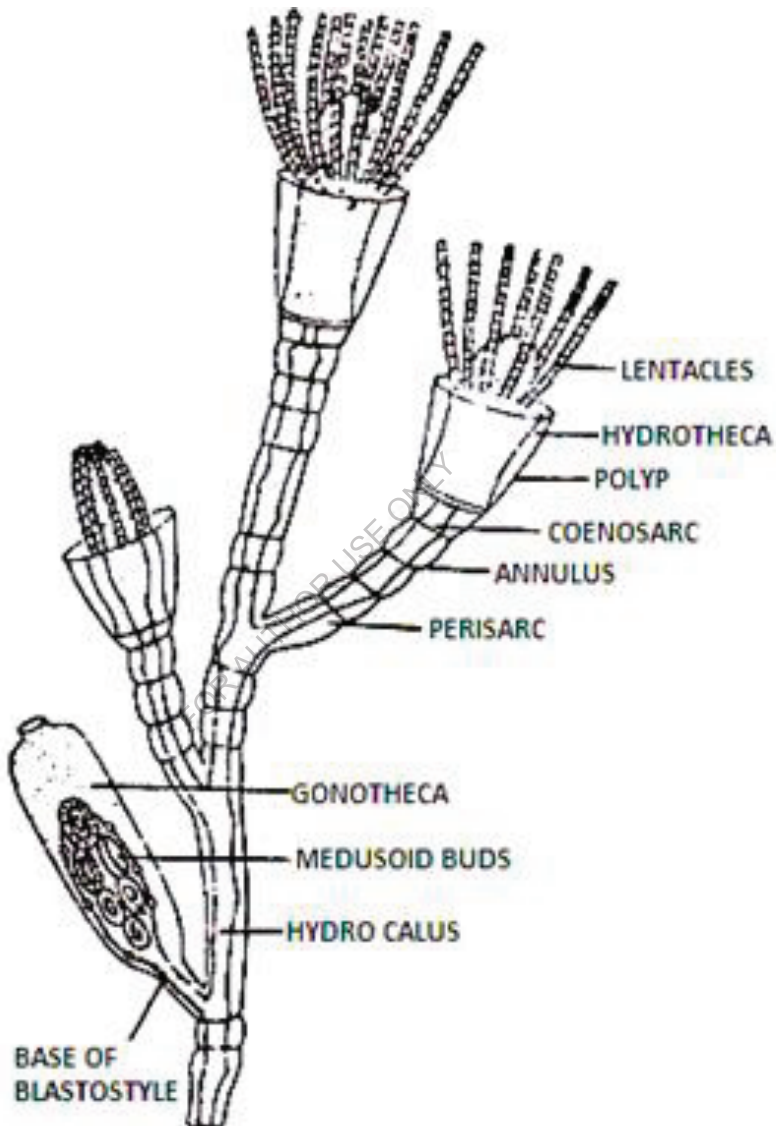
SYCON SPONGE



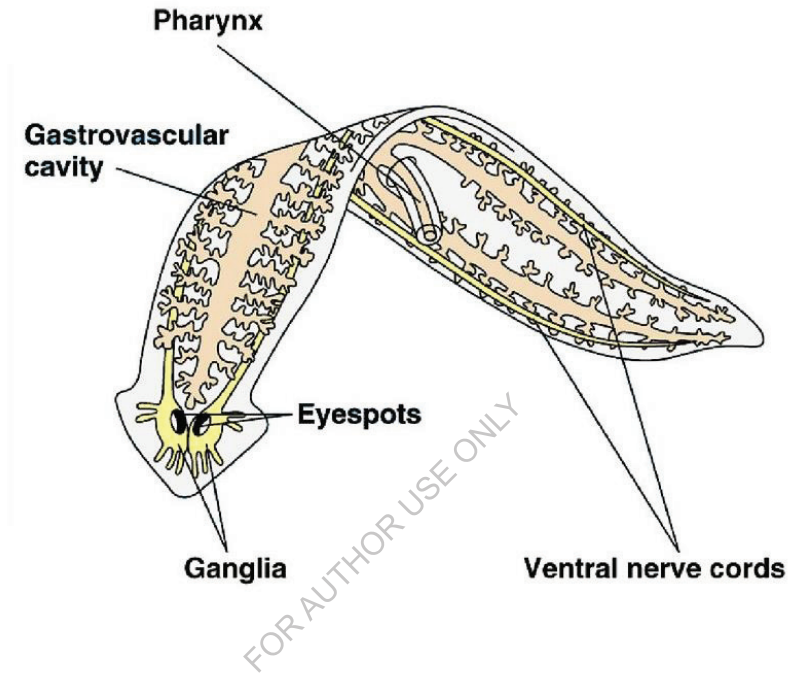
AURELIA



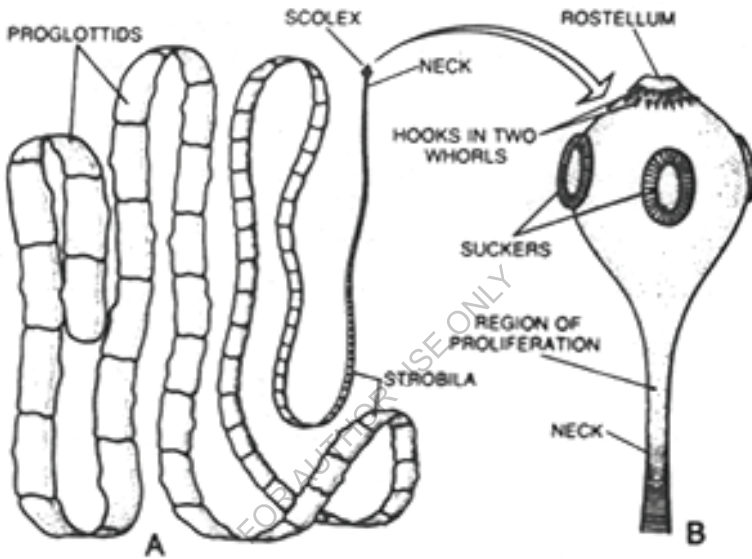
OBELIA



PLANARIA

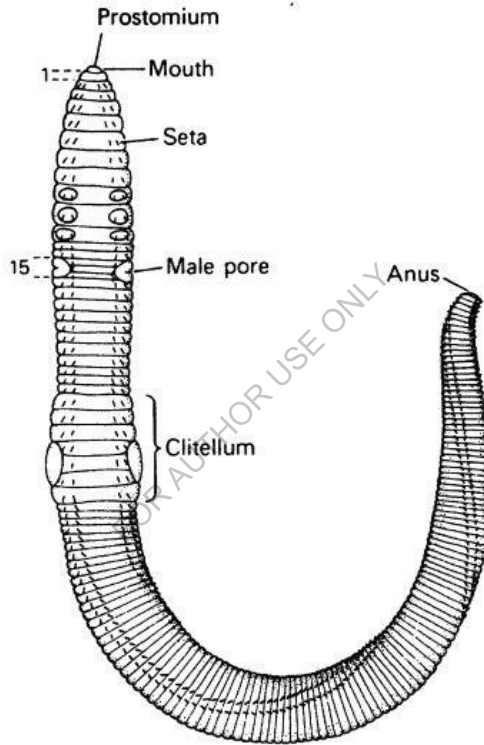


TAPE WORM



Taenia solium : A. whole; B. its scolex

EARTH WORM



NEREIS

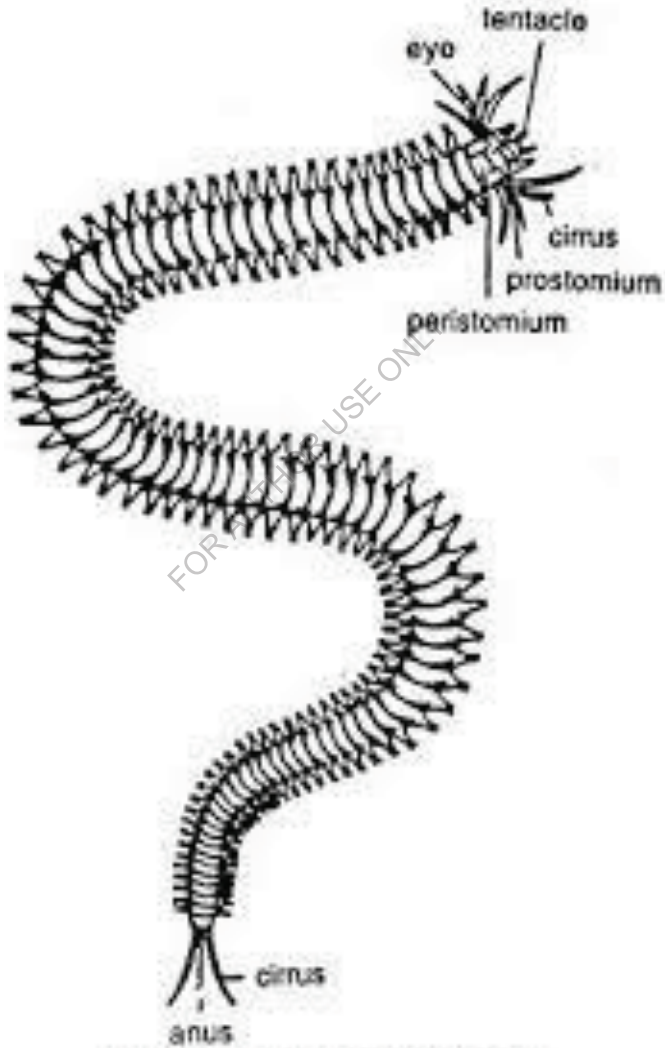
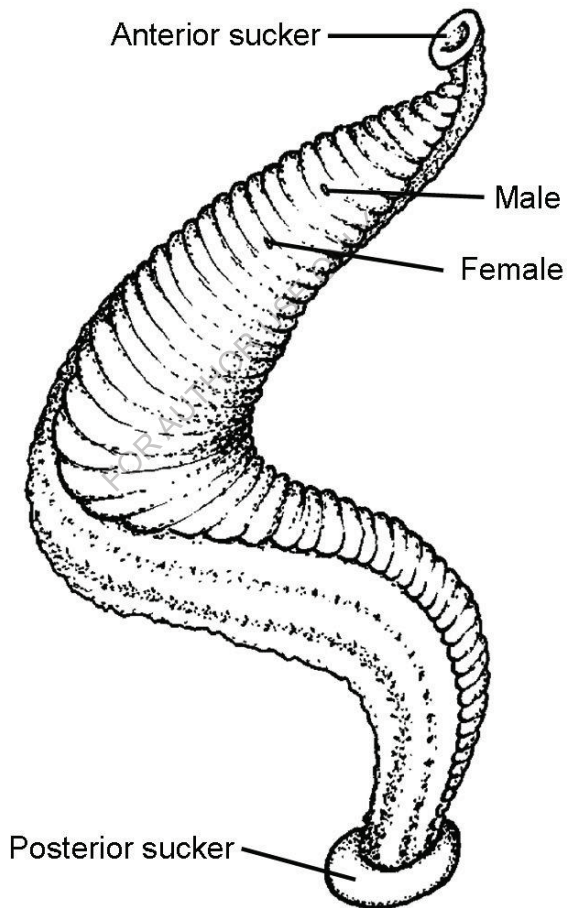
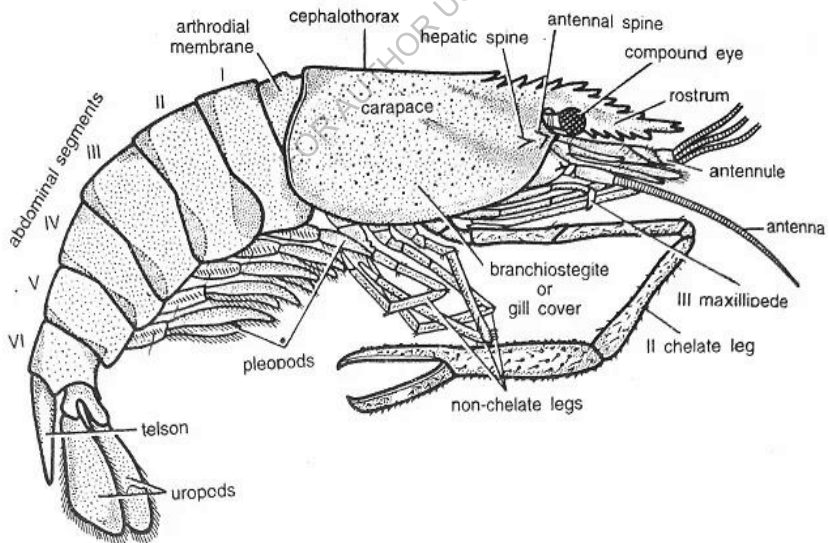


Fig. 24.1. *Nereis (Nereis) chilkaensis*

LEECH



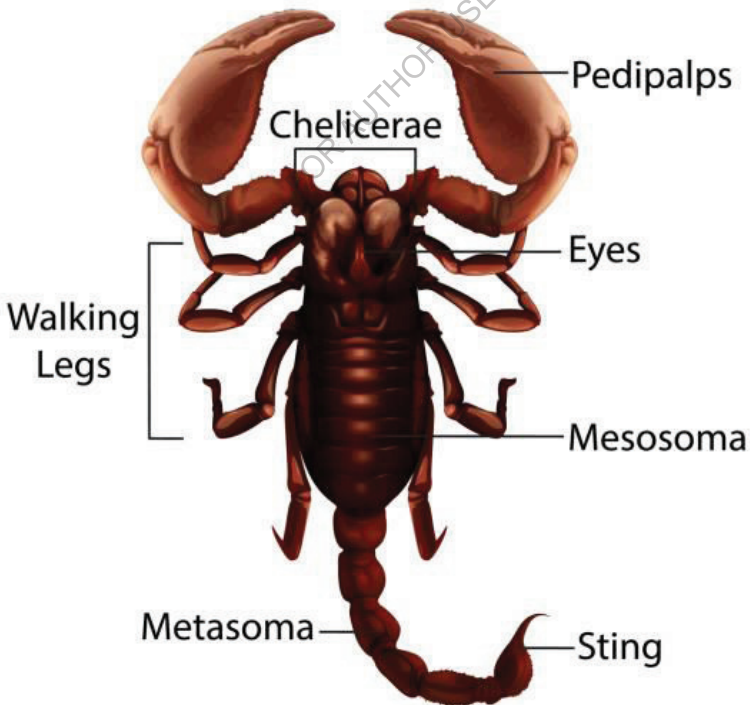
PRAWN



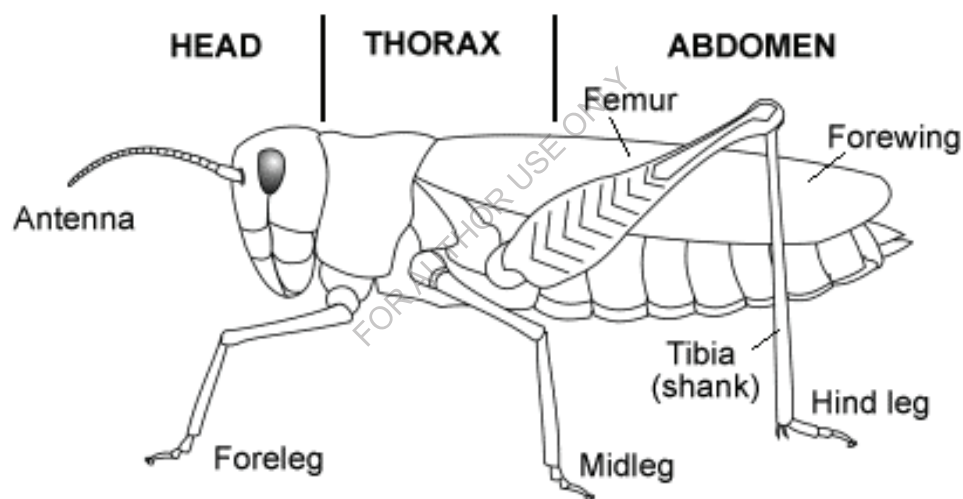
Palaemon.

SCORPIAN

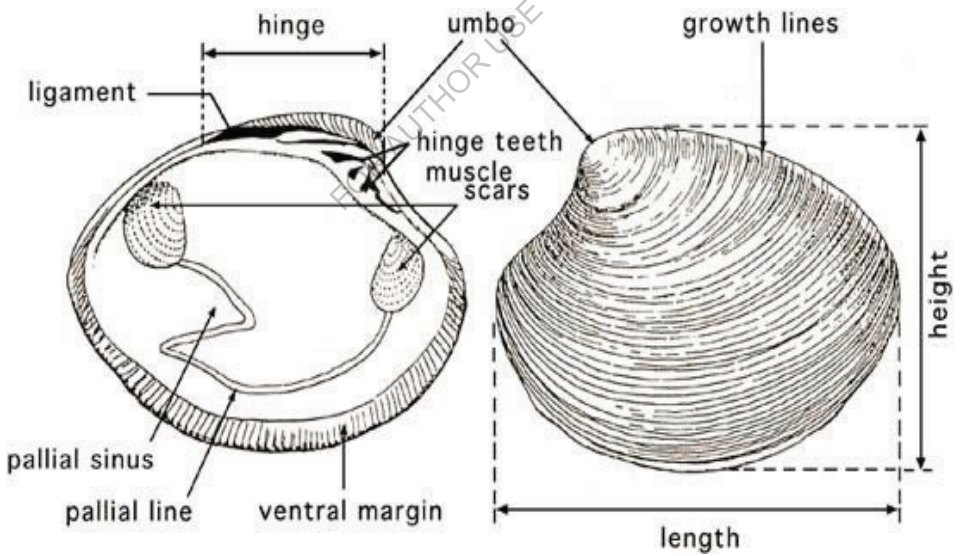
External Anatomy of a Scorpion



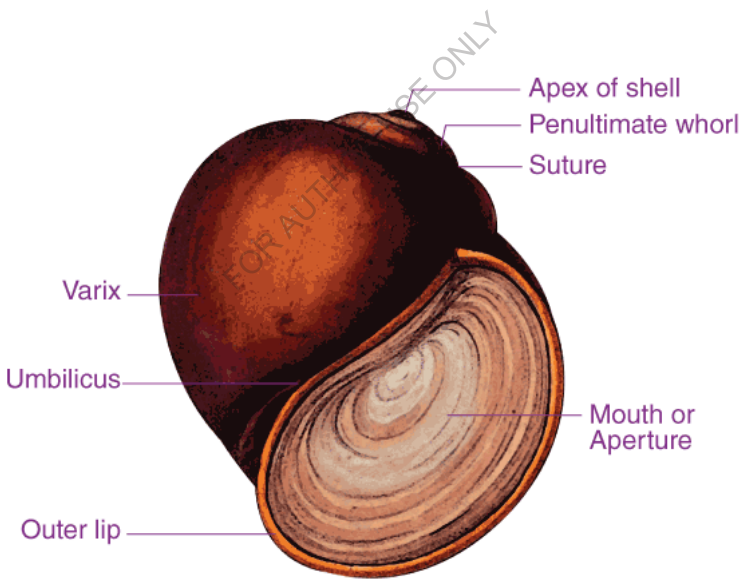
GRASS HOPPER



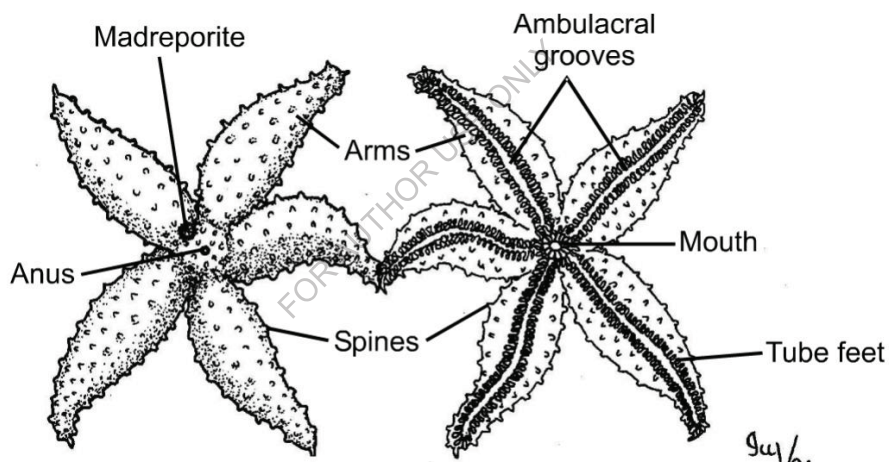
FRESH WATER MUSSEL



PILA

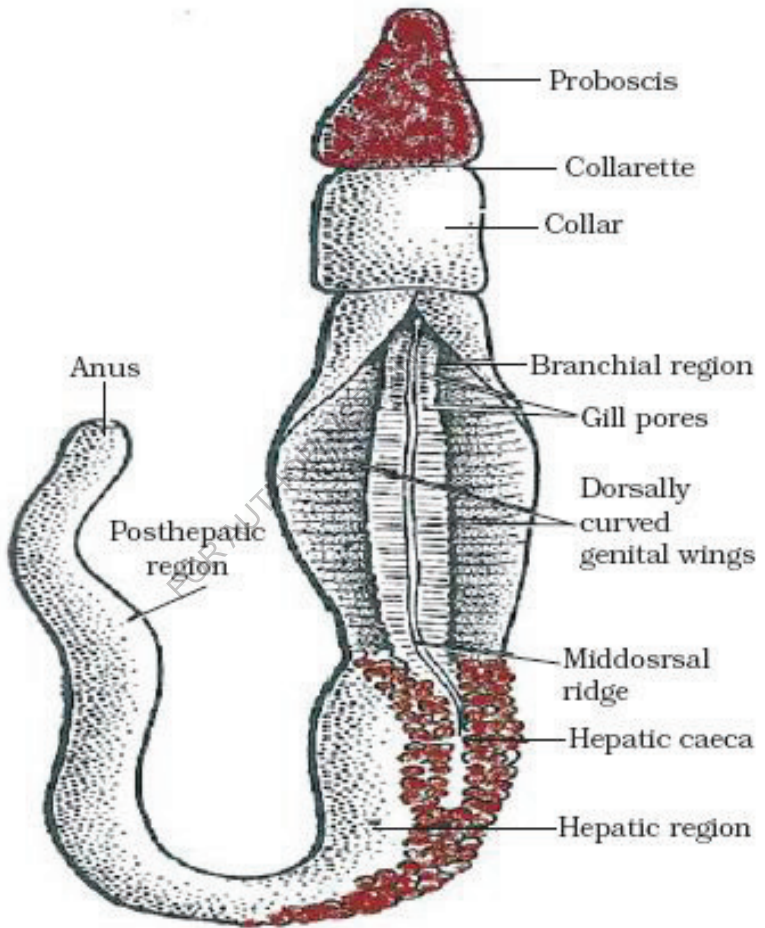


STAR FISH

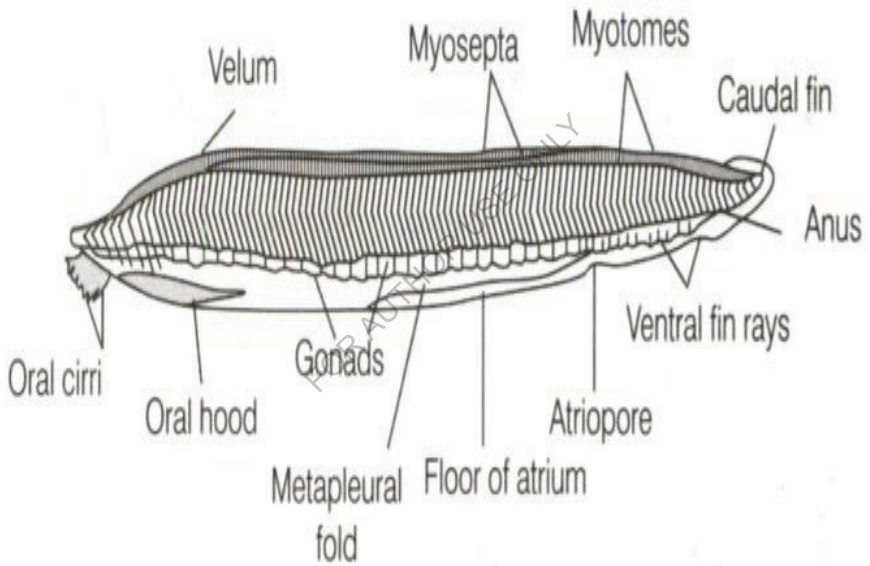


PROTO CHORDATA

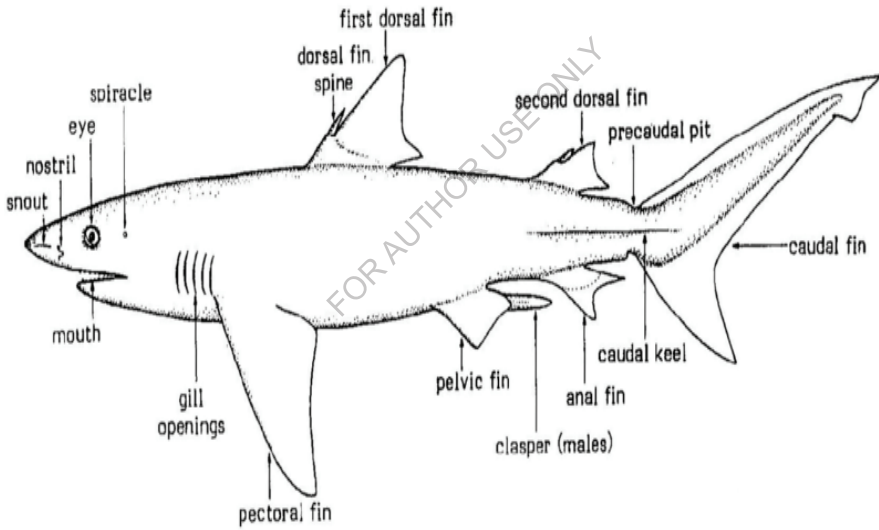
(BALANOGLOSSUS)



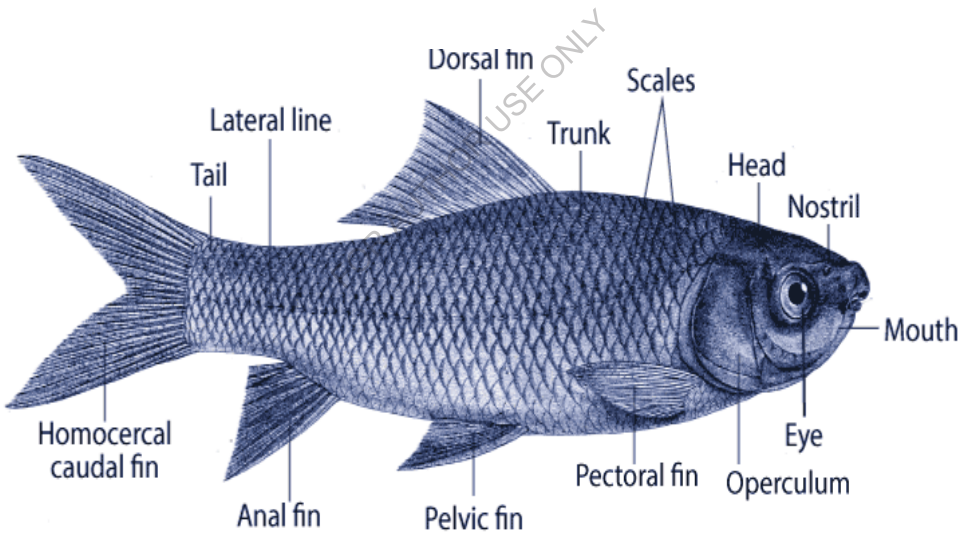
AMPHIOXUS



SHARK



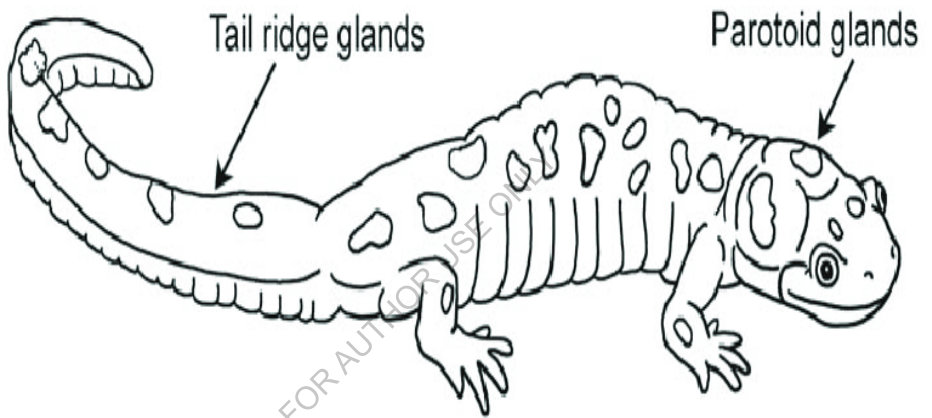
CATLA



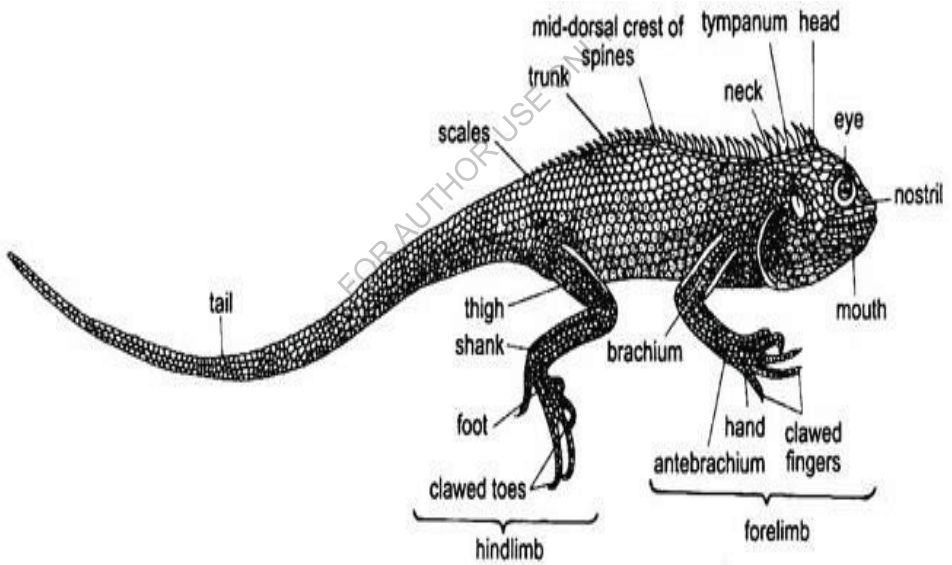
FROG



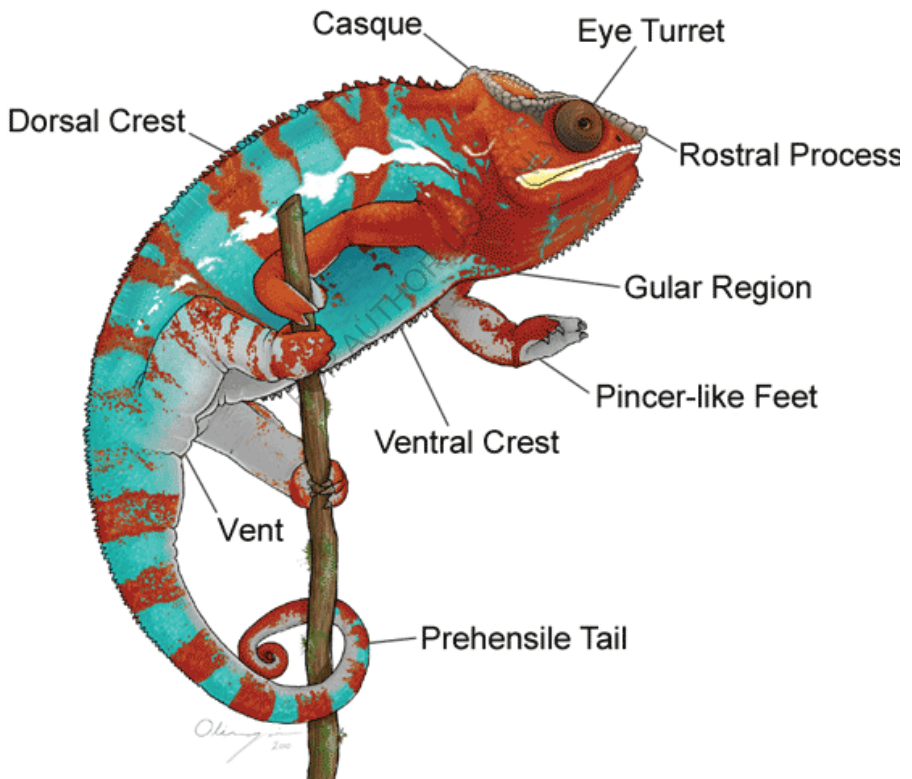
SALAMANDER



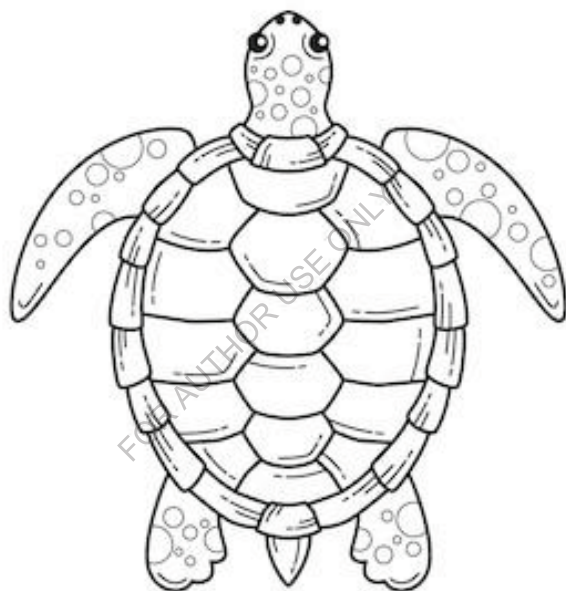
CALOTES



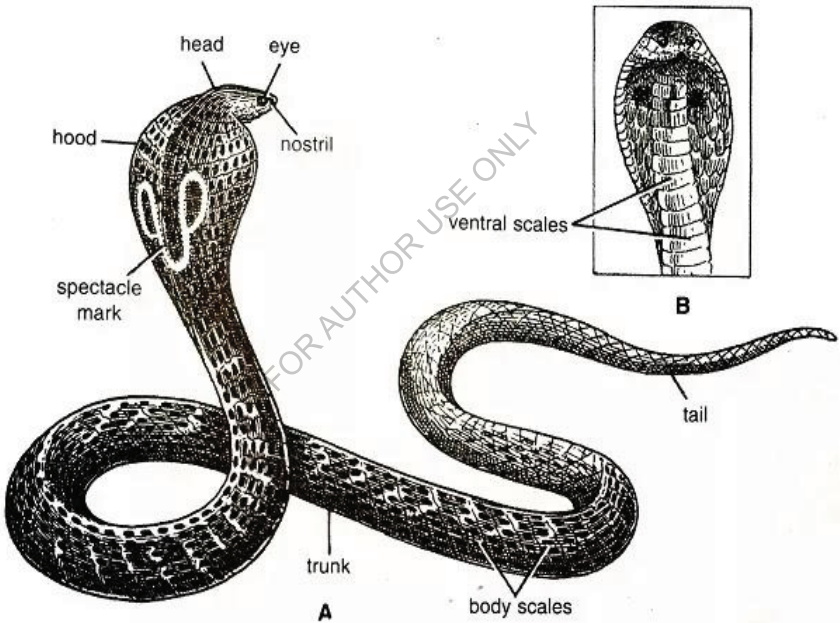
CHAMELEON



TURTLE

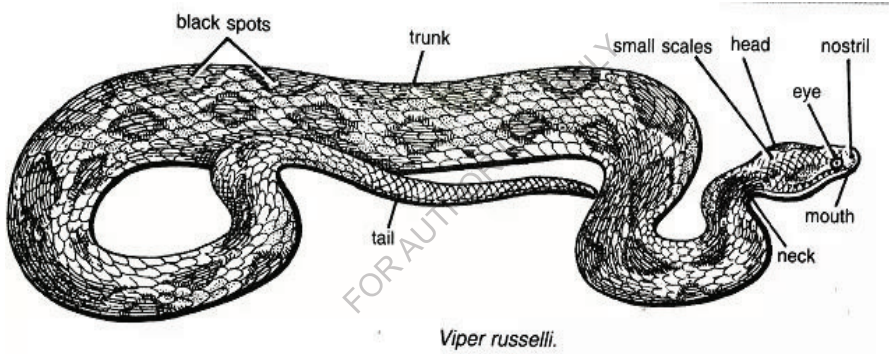


COBRA

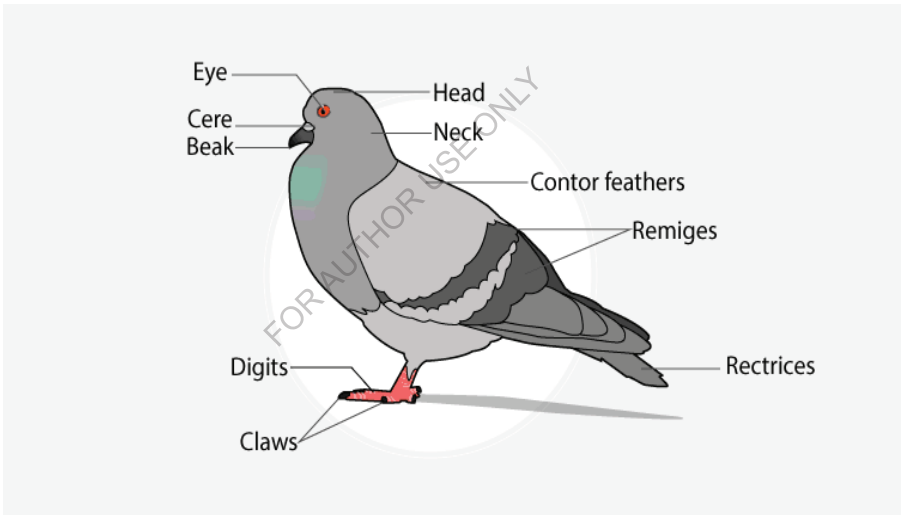


Naja naja. A — Hood expanded; B — Ventral view of hood.

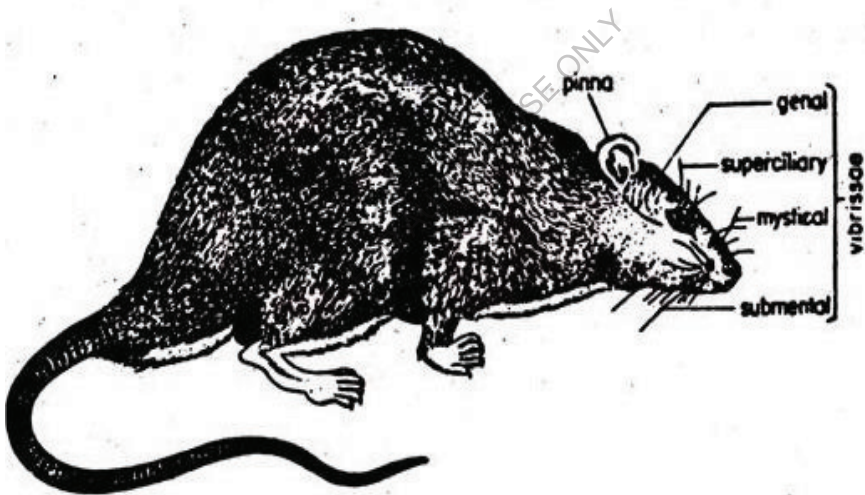
VIPER



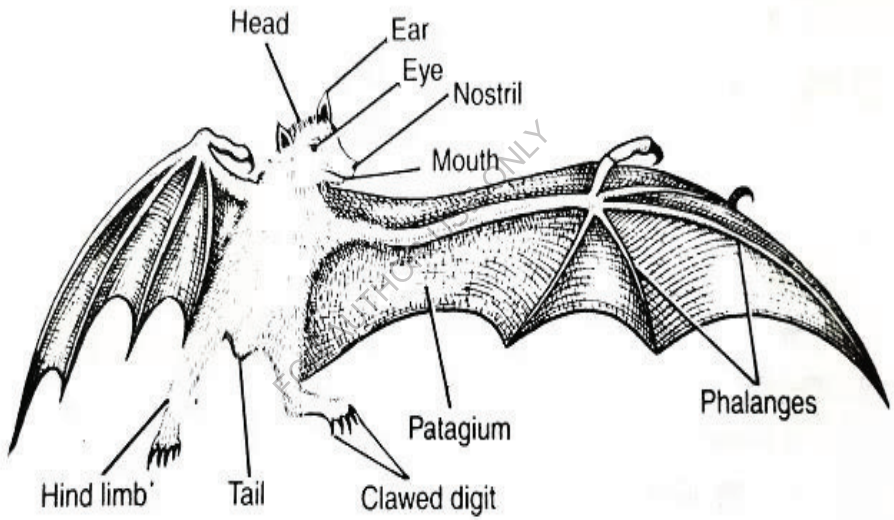
PIGEON



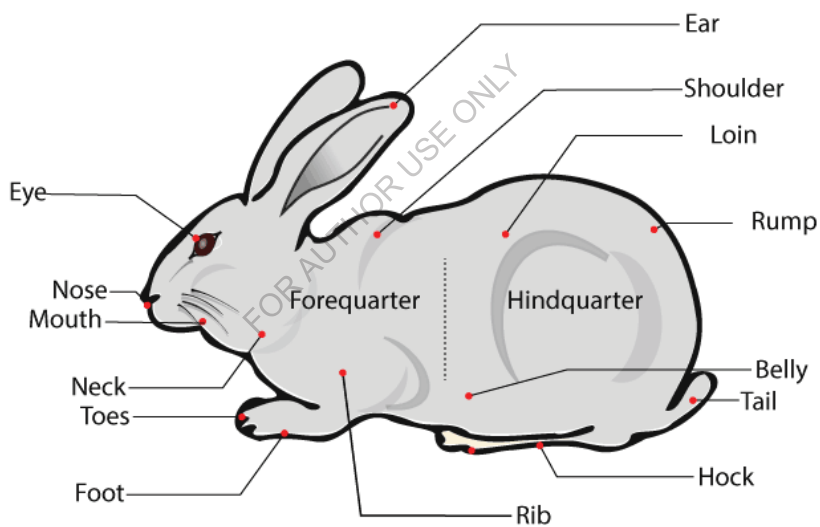
RAT



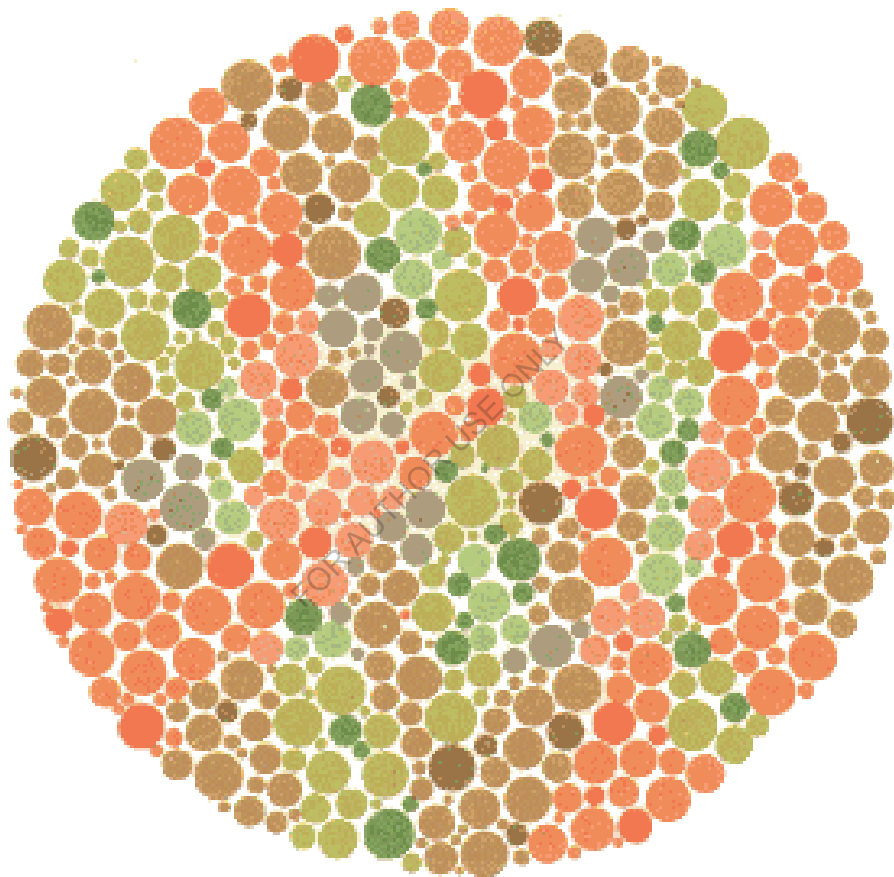
BAT



RABBIT



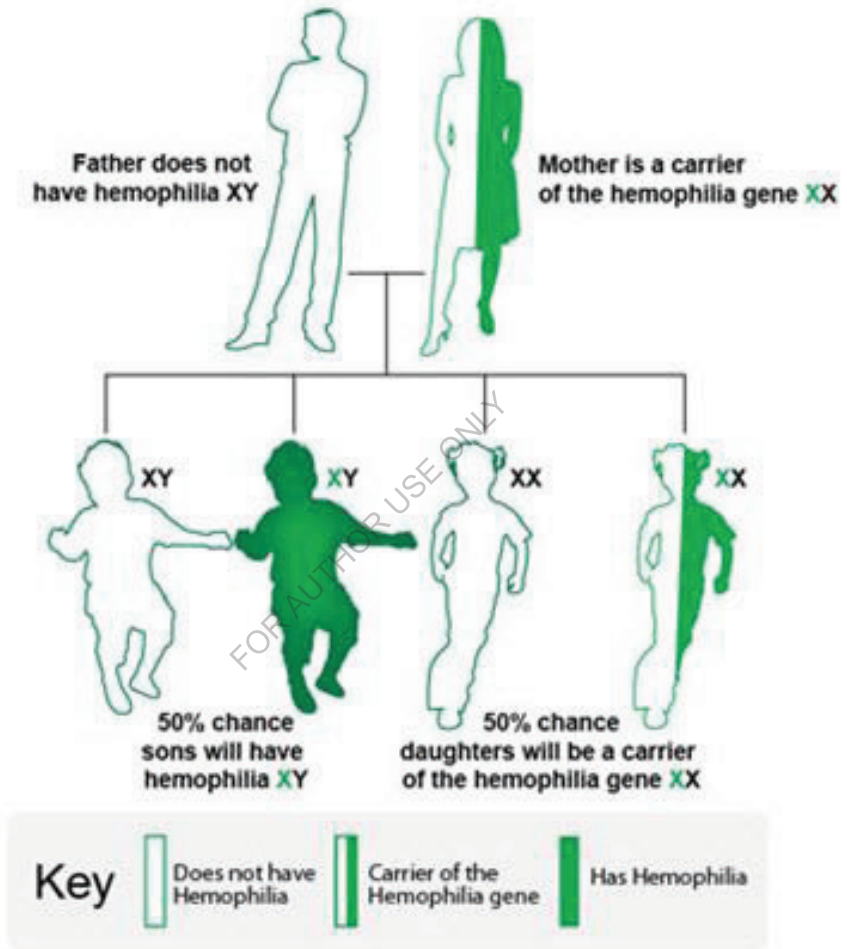
COLOUR BLINDNESS



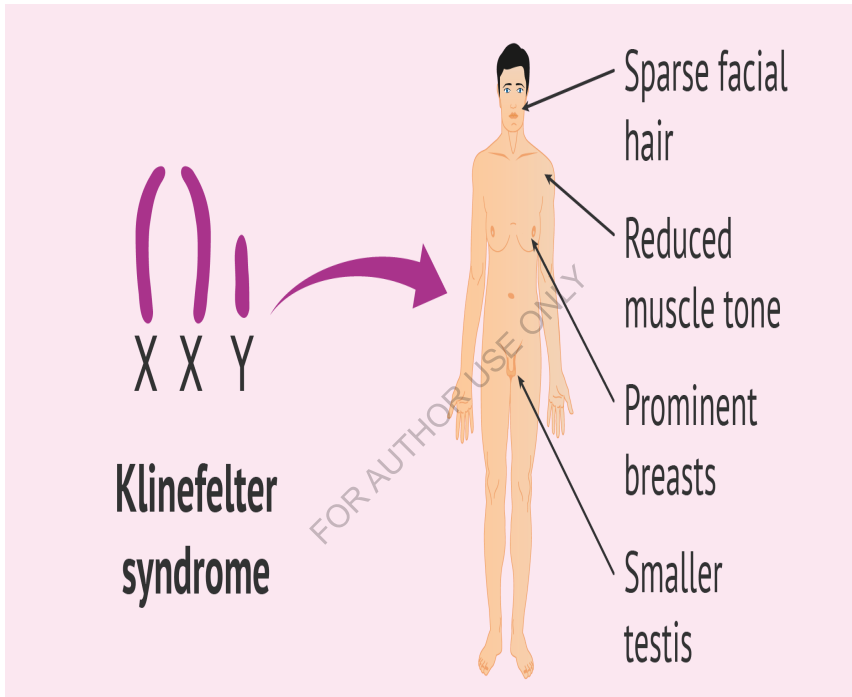
DOWNS SYNDROME



HAEMOPHILIA



KLINEFELTERS SYNDROME



IV.GENERAL LABORATORY SAFETY

PRECAUTION

6.1. DO AND DON'T LABORATORY SAFETY:

Safety in the Laboratory Should always be in your mind throughout this manual. Safety recommendations are given, below are some general considerations that anyone in a laboratory should know.

- Follow all instructions carefully. Use special care when you see the word caution in your laboratory instructions. Follow the safety instructions given by your teacher.
- Determine the location of Fire Extinguishers, Chemical safety showers and Eye washers, Chemical Spill Kits and alternative exit routes for lab evacuation.
- Remember that smoking, eating or drinking in the lab room is totally prohibited.
- Wear lab aprons when working with chemicals, hot material or preserved specimens.
- Wear safety goggles when using dangerous chemicals, hot liquids or burners.
- Any chemicals spilled on the hands or other parts of the skin should be washed off immediately with a plenty of running water.
- If you have an open skin wound, be sure that it is covered with a water proof bandage.
- Never work alone in the laboratory.
- Keep your work area clean and dry.
- Turn off all electrical equipment, water and gas when it is not in use, especially at the end of the laboratory period.
- Tie back long hair.
- Report all chemicals spills or fluids to your instructor immediately for proper clean up.

SPECIAL PRECAUTIONS FOR WORKING WITH HEAT OR FIRE

- Never leave a lighted Bunsen burner or hot object unattended. When an object is removed from the heat & left to cool, it should be placed where it is shielded from contact.

- Inflammable liquid bottles should not be left open, not dispensed near a naked flame, hot electric element or electric motor.
- Use test tube holders to handle hot laboratory equipments.
- When you are heating something in a container such as a test tube, always point the open end of the container away from yourself & others.
- Use only Pyrex glassware's for heating.
- Allow hot materials to cool before moving them from your lab station.
- Make sure that Bunsen burner hoses fit tightly.

SPECIAL PRECAUTIONS FOR WORKING WITH CHEMICALS

1. Never taste or touch substances in the laboratory without specific instructions.
2. Never smell substances in the laboratory without specific instructions.
3. Use materials only from containers that are properly labeled.
4. Wash your hand after working with chemicals.
5. Do not add water to acid. Instead, dilute the acid by adding it to water.
6. Mix heat generating chemicals slowly.

SPECIAL PRECAUTIONS FOR WORKING WITH ELECTRICAL EQUIPMENT:

1. Make sure the area under & around the electrical equipment is dry.
2. Never touch electrical equipment with wet hands.
3. Make sure the area surrounding the electrical equipment is free of flammable materials.
4. Turn off all power switches before plugging an appliance into an outlet.

SPECIAL PRECAUTION FOR WORKING WITH GLASSWARE'S AND OTHER LABORATORY EQUIPMENTS:

1. Become familiar with the names and appearance of all the laboratory equipments you will use.
2. Never use broken or chipped glassware.
3. Make sure that all glassware's are clean before you using it.
4. Do not pick up broken glass with your bare hands. Use a pan and a brush.
5. If a Mercury thermometer breaks, do not touch the mercury. Notify your teacher immediately.
6. Do not aim the mirror of your microscope directly at the sun.
7. Direct sun light can damage the eyes.
8. Use care handling all sharp equipments, such as scalpels and dissecting needles.

SPECIAL PRECAUTIONS FOR WORKING WITH LIVE OR PRESERVED SPECIMENS:

1. If live animals are used treat them gently. Follow instructions for their proper care.
2. Always wash your hands after working with live or preserved organisms.
3. Specimens for dissection should be properly mounted and supported. Do not try to cut a specimen while holding it in the air.
4. Do not open Petri dishes containing live cultures unless you are directed to do so.
5. Detergents (Detol 5 – 10%) should be used to sterilize and clean benches, glassware and equipment.
6. Safety cabinet should be used while working with microbes.
7. Lab coats should be worn during the work in the lab.
8. Disposable items should be collected and autoclaved.

6.2. FIRST AID AND SAFETY MEASURES

1. Injuries: bleeding should be reduced using bandages; the wound
2. Should be cleaned with iodine alcohol mixture and wrapped with sterile bandage.
3. Acid and fire burns: body burns must be washed immediately with tap water. Eye burns must be washed using eye washer, special cream for burns can be used.
4. Poisoning: if any toxic chemical is swallowed, the mouth must be rinsed with water, in case of acid, milk is drunk, in case of alkaline, diluted acetic acid (vinegar) can be used.
5. Skin contamination requires washing with water and removal of contaminated clothing, if the contaminant is insoluble in water remove with soap and water.



RESPONSIBILITY

The laboratory instructor and laboratory aides are responsible for maintaining this procedure.

PROCEDURE

- Laboratory procedures all employees working within the district laboratories are provided with a copy of applicable laboratory safety system procedures (chemical hygiene plan) and provided with training necessary for safe operations.
- Work study students are also provided with safety training.
- All employees are required to familiarize themselves with the procedures and sign a statement acknowledging receipt and understanding of the procedures.
- The signed statement is kept on file by the chemical hygiene officer.
- Laboratories are used for laboratory activities only.
- If you make a mess, you are responsible for ensuring it is cleaned up.
- Disinfectant is used to wipe lab working areas before and after lab activity.
- Good laboratory practices are to be followed at all times. if a student is in doubt, direction of the instructor or laboratory aide should be sought before proceeding in any activity.
- All laboratory assignments must take place and be completed in the laboratory. No lab activity is allowed in classrooms.
- All labs must be completed on the assigned date, no; open labs are permissible unless an instructor or lab aide is present.

- Students must not be allowed to congregate in the halls outside the lab while waiting for practical exams to start. this is a fire hazard and disruptive to other classes. Give students a definite time to come to the lab to begin the practical.
- Personal protective equipment safety glasses must be worn during all laboratory activity involving chemicals, boiling water, use of the Bunsen burner and dissecting by students, instructors, and laboratory aides.
- Other personal protective equipment (i.e. gloves) is to be worn as directed by laboratory setup instructions or the instructor.
- Failure to wear appropriate personal protective equipment in laboratory situations will lead to disciplinary measures, up to and including dismissal from employment or expulsion from the college program.
- Personnel working or using the lab must confine long hair, and wear regular length lab coats. when applicable, wear gloves to protect hands from contamination inspect gloves before use.
- No clogs, flip-flops, open-toed shoes, sling back heels, sandals, or bare feet are allowed in lab. Regular length lab coats must be worn at all times over all clothing.
- Know the location of all fire and first aid equipment.
- Use of the lab students is never allowed in the laboratory areas without supervision of the instructor or laboratory aide.
- Coats, purses, extra books, or any other materials not being used in the current laboratory activity are not allowed in the working area. These items are to be placed in the designated area during laboratory times.
- The doors to the laboratory and chemical storage areas must be locked upon leaving by the instructor and/or laboratory aide.
- Damaged or defective equipment all equipment and materials must be kept in its designated storage area when not in use. Inspect equipment and materials before use. Never used damaged or defective equipment.
- Discard in the appropriate waste receptacle, remove from the work area and place a written notice on it of suspected damage or defect to prevent possible use, and • report damaged or defective equipment to the laboratory aide .
- No food, drink, smoking, or children are allowed in the college laboratory areas at any time.
- Never walk away from Bunsen burners when in use.
- Be alert for unusual or unsafe circumstances in the laboratory areas. report all concerns immediately to the laboratory aide. if you make a mess, you are responsible for ensuring it is cleaned.

6.3. HANDLING EQUIPMENT AND MATERIALS

- Properly label all glassware and slides.
- Keep all equipment and materials away from the edge of the lab benches.
- Use protective aids (e.g. test-tube clamps, hot pads) when handling hot glassware.
- Light gas burners only as instructed. Be sure no volatile or flammable materials are being used nearby. Remember to have long hair tied back, no dangling jewelry or loose clothing
- When heating a test tube do not look into the mouth of the tube or point it into the direction of any person.
- Heat sources should only be on when in use. Do not leave heat sources unattended. When in doubt, students should assume a heat source is on or hot.
- Operate electrical equipment in a dry area with dry hands.
- When removing an electrical plug from its socket, pull the plug, not the cord.
- Inform your instructor immediately if equipment or materials are broken or not working properly.
- Never return reagents back into their original containers. Keep containers capped when not in use.

6.4. HANDLING HAZARDOUS MATERIAL

- Wear appropriate eye protection as directed by the instructor.
- Wear disposable gloves when handling hazardous materials including chemicals, organisms, or bodily fluids.
- Wear protective clothing when required by the instructor.
- Follow correct disposal procedures as detailed in “Clean up and Disposal.”
- To prevent contamination by needle stick injuries, always use new needles and lancets.
- Use the fume hood whenever noxious, corrosive or toxic fumes are produced or released.

6.4. HANDLING HAZARDOUS MATERIALS BACTERIA AND FUNGI

1. Never open Petri dishes containing bacterial or fungal growth unless directed to do so by the instructor.

2. Dispose of all discarded bacterial and fungal cultures as directed by the instructor.

CLEANUP AND DISPOSAL

1. Follow correct disposal procedures dispose of used materials promptly and in the appropriate disposal containers.
 - a. Dispose of broken glass in labeled glass disposal containers.
 - b. Dispose of needles and lancets in puncture-proof sharp containers
 - c. Dispose of used chemicals in labeled and capped waste containers as directed by your instructor.
 - d. For disposal of items contaminated with organisms or body fluids,
 - Decontaminate slides, spectrophotometer tubes, and other glassware in 10% bleach
 - Dispose of soft items (i.e. gauze or swabs) in labeled bio hazard containers.
 - Dispose of all animal tissues in labeled bio hazard containers.
2. Place micropipette tips, sample vials, and contaminated glassware in a labeled waste container or designated area.
3. See instructor for disposal of cadaver tissues.

ALLIED ZOOLOGY PRACTICAL -I

MAJOR PRACTICAL

- a. Cockroach/Fish –Digestive.
- b. Qualitative detection of excretory products (Ammonia, Urea, Uric acid).

MINOR PRACTICAL

- a. Mouth parts of Honey Bee, Mosquito.
- b. Fish – cycloid scale, ctenoid scale and placoid scale.
- c. ABO blood group.

SPOTTERS

Amoeba, Paramecium, Trypanosoma, Euglena, Plasmodium, Leucosolenia, Sycon sponge, Aurelia, Obelia, planaria, Liver fluke, Tapeworm, Cockroach, Planaria, Earthworm, Nereis, Leech, Prawn/Shrimp, Scorpion, Grasshopper, Fresh water mussel, Pila, Starfish. Protochordata and Vertebrata – Amphioxus, Shark, Catla, Frog, Salamander, Calotes, Chamaeleon, Turtle, Cobra, Viper, Pigeon, Rat, Bat, Rabbit. Colour Blindness, Haemophilia, Klinefelter's syndrome, Down's syndrome.

REFERENCE

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